



Is it alpha or beta? Decomposing hedge fund returns when models are misspecified

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ABSTRACT

We develop a novel approach to separate alpha and beta under model misspecification. It comes with formal tests to identify less misspecified models and sharpen the return decomposition of individual funds. Our hedge fund analysis reveals that: (i) prominent models are as misspecified as the CAPM, (ii) several factors (time-series momentum, variance, carry) capture alternative strategies and lower performance in all investment categories, (iii) fund heterogeneity in alpha and beta is large—an important result for fund selection and models of active management, (iv) performance is increasingly similar to mutual funds, (v) fund valuation is sensitive to investor sophistication.

1. Introduction

The average return that any hedge fund i delivers to investors is the sum of two components. The alpha component ac_i^* is based on private information—it captures the net return that the fund produces by using its proprietary information signals. The beta component bc_i^* is based on public information—it captures the return that the fund produces by following mechanical trading strategies. The economic importance of these two components is potentially large. A commonly held view is that hedge fund managers deliver positive alphas because they are more sophisticated, less constrained, and more incentivized than mutual fund managers. In addition, the literature consistently emphasizes that hedge funds take advantage of the positive returns of alternative strategies that

are weakly correlated with the equity market (e.g., Carhart et al., 2014; Pedersen, 2015).

Decomposing returns is key for evaluating the performance and risk profile of hedge funds. It is therefore important for researchers, investors, and policymakers alike. This decomposition is likely to exhibit substantial variation across funds as they follow many investment strategies and private information signals (e.g., Lhabitant, 2007). As a result, averaging across funds provides limited information about the entire fund population. For example, it does not reveal how many funds deliver positive alphas—an important quantity for testing the predictions of equilibrium models of active asset management. Capturing fund heterogeneity is also crucial for hedge fund investors because they can only invest in a handful of funds (Bollen et al., 2021). These arguments

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call for the estimation of the entire distributions of the alpha and beta components characterized by the densities ϕ_{ac}^* and ϕ_{bc}^* .

The estimation of these distributions is hampered by misspecification. Capturing all the strategies followed by hedge funds is notoriously difficult, which implies that any chosen model k is likely misspecified—that is, it omits relevant factors that drive the beta components of individual funds. In this case, we cannot infer the true components ac_i^* and bc_i^* . Instead, we can only observe the estimated components $\hat{ac}_i^k$ and $\hat{bc}_i^k$, which serve as inputs to compute the distributions $\hat{\phi}_{ac}^k$ and $\hat{\phi}_{bc}^k$. These distributions are imperfect and noisy versions of ϕ_{ac}^* and ϕ_{bc}^* . They are imperfect because the alphas absorb the average returns of the factors omitted by the model. Put differently, high alphas are simply hidden betas. The distributions are also noisy because of the sampling variation of the omitted factors. This misspecification-driven variation, which affects all funds, does not vanish even when the fund population grows large.

We propose a new methodology to address the challenges of misspecification. We develop a nonparametric approach for estimating the distributions ϕ_{ac}^k and ϕ_{bc}^k for any chosen factor model. Its key feature is to explicitly account for the dual impact of misspecification. First, it provides a framework for comparing models and identifying the ones with greater ability to capture alternative strategies. Using models less prone to misspecification produces a sharper identification of the true alpha and beta components ac_i^* and bc_i^* . Second, it comes with a full-fledged asymptotic theory that incorporates the estimation noise due to misspecification. This contribution is crucial to conduct proper statistical inference on the fund alphas and betas and perform valid model comparison tests.

Our approach contributes to previous studies which measure the distribution of fund alphas under correct specification (e.g., Barras et al., 2022; Harvey and Liu, 2018). In contrast, we account for the impact of misspecification, which is essential for decomposing hedge fund returns. Our approach also provides the first formal comparison of misspecified models in a large population of funds. It therefore departs from previous tests based on a small number of assets (e.g., Kan and Robotti, 2009; Kan et al., 2013). We cannot apply such tests here because they require the inversion of the entire return covariance matrix—an operation that cannot be performed because the number of funds is larger than the number of return observations.

To demonstrate the benefits of our approach, we apply it to a set of nine diverse models. In addition to the CAPM, we include four standard models commonly used in previous work: the models of Carhart (1997), Fama and French (2015), Fung and Hsieh (2004), and Asness et al. (2013). We examine the two machine learning models of Kozak et al. (2020) trained on 50 characteristic-based equity portfolios. Finally, we consider two models formed with five additional factors that plausibly capture hedge fund strategies: marketwide illiquidity, betting-against-beta (BAB), variance (short position), carry, and time-series (TS) momentum.¹ The first model by Joenväärä et al. (2021) (JKKT) is a Carhart model with illiquidity, BAB, and TS momentum. Based on the work by Carhart et al. (2014) and Pedersen (2015), we form a second model (CP) that includes all five additional factors (alongside with market and size).

We conduct our analysis between 1994 and 2020 using monthly data on 5,231 hedge funds. Following Joenväärä et al. (2021), we carefully aggregate four different databases and mitigate the various biases that affect hedge fund reporting (backfill, selectivity, and survivorship). For each model k , the estimation of the alpha and beta distributions ϕ_{ac}^k and ϕ_{bc}^k requires as inputs the estimated components $\hat{ac}_i^k$ and $\hat{bc}_i^k$ for each fund. To this end, we run a regression of the excess return (net of fees)

of each fund on the excess returns of the factors included in model k . We then compute the two components as $\hat{ac}_i^k = \hat{\alpha}_i^k$ and $\hat{bc}_i^k = \hat{\mu}_i - \hat{ac}_i^k$, where $\hat{\alpha}_i^k$ and $\hat{\mu}_i$ are the estimated alpha and average return of the fund.

Our formal comparisons uncover sharp differences between models. In our baseline analysis, we use as reference the CAPM—a simple model that is ill-equipped to capture hedge fund returns (beyond the market premium). If a proposed model delivers the same alphas as the CAPM, it is therefore unable to capture any of the alternative strategies followed by hedge funds. We reach this conclusion for all the standard and machine learning models. In contrast, the JKKT and CP models are able to capture the premia on alternative factors. In particular, the difference between the CP model and the CAPM is statistically and economically large, stable over time, and robust to the inclusion of factor trading costs and alternative filters for reducing data biases.

These results have strong implications for the fund return decomposition. Consistent with previous studies, the standard models produce large alpha components—more than 70% of the funds have a positive alpha equal to 2.7% per year on average.² At the same time, the average beta component due to the non-market factors (i.e., all factors but the market) barely reaches 0.2% per year. Therefore, hedge funds deliver superior performance while being immune to alternative sources of risk. The CP model reverses this conclusion. We find that only 50% of the funds deliver positive alphas. The average alpha drops to 0.4% per year, whereas the non-market factors become the main contributors to average returns (2.9% per year).

The five additional factors are economically important. The majority of the funds load positively on each of them, which supports the view that hedge funds follow alternative strategies to boost returns (e.g., Carhart et al., 2014). We also find that underperforming funds are particularly aggressive in their factor exposures—possibly to boost returns and hide their lack of skills. Out of the five factors, the CP model includes the three most relevant ones—TS momentum, variance, and carry—whose respective return contributions are equal to 1.1%, 0.8%, and 0.4% per year. This finding explains the edge of the CP model over the JKKT model. It is also in line with ample anecdotal evidence that hedge funds follow trends, take short option positions, and buy cheap assets with high carry (Asness et al., 2015; Pedersen, 2015).

Next, we turn to the analysis of three diverse investment categories: (i) equity funds, (ii) macro funds, and (iii) arbitrage funds. We find that the superiority of the CP model applies to each category. The average alpha is equal to 0.6%, -0.4%, and 0.9% per year for equity, macro, and arbitrage funds (versus 2.4%, 3.7%, and 2.8% under the CAPM). The widespread reduction in alphas arises because some additional factors transcend style boundaries. For instance, carry is widely used by hedge funds regardless of their styles when they implement currency carry trades, buy high-yield bonds, or favor value stocks (Pedersen, 2015). At the same time, some factors primarily matter for specific styles. For instance, TS momentum is crucial for capturing trend-following strategies in the macro category. This result highlights the potential benefits of including style-specific factors to lower model misspecification.

The analysis of investment styles reveals a large fund heterogeneity. Under the CP model, the cross-sectional volatilities in the alpha and beta components range between 4.9% and 11.0% per year. Therefore, sorting funds based on the three categories does not produce homogeneous groups with similar performance and risk profiles. This heterogeneity provides key insights on how to form models using style information. It invalidates the common practice of peer benchmarking which imposes that all funds in a given category are benchmarked against the same style index. It also suggests that style-specific factor

¹ As discussed in Section 4.2, the illiquidity, BAB, carry, and TS momentum factors are constructed by Pástor and Stambaugh (2003), Frazzini and Pedersen (2014), Kojien et al. (2018), and Moskowitz et al. (2012).

² A non-exhaustive list of papers that document positive average alphas under the standard models includes Avramov et al. (2013), Buraschi et al. (2014b), Capocci and Hübner (2004), Chen et al. (2017), Diez de los Rios and García (2010), Kosowski et al. (2007).

models are more effective when applied to narrowly-defined investment categories (see, for instance, Duarte et al., 2006).

Fund heterogeneity introduces uncertainty in the fund selection process. To reduce this uncertainty, we examine whether fund characteristics that proxy for managerial incentives and flexibility (e.g., performance fees, lockup period) can be used as initial filters. We find that it is the case—the proportion of positive-alpha funds is systematically higher among funds with higher incentives/flexibility. Fund heterogeneity has also implications for models of active management. The Berk and Green (2004) model predicts that all funds deliver zero alphas. Whereas this prediction holds quite well for mutual funds, it is at odds with our empirical results. In contrast, the model of Gârleanu and Pedersen (2018) is consistent with fund heterogeneity as investors are compensated for search costs—an intuitive explanation given the complex process of evaluating hedge funds.

Our analysis uncovers notable time trends between 1994 and 2020. Whereas hedge funds remain unique in their exposures to alternative strategies, they converge towards mutual funds along two dimensions. First, their performance becomes increasingly similar as the gap in alphas reaches its lowest value of 1.6% in 2020. Second, they load increasingly on the equity market after the 2008 crisis. The equity market is now the most prevalent risk factor—its average contribution is equal to 2.3% per year, which represents 44% of the total beta component.

Finally, our approach sheds light on the impact of investor sophistication on hedge fund valuation. Whereas the CP alpha gives the valuation of a hypothetical investor able to replicate the five additional factors, the CAPM alpha determines the valuation of a less sophisticated investor who can only invest in the market. We find that the average valuation gap is large (2.6% per year) as the CAPM investor values the alpha component as well as the return contribution of alternative strategies she cannot replicate. Examining fund flows, we find that real-world hedge fund investors are closer to the CP investor in terms of sophistication. Decomposing returns across funds with low and high flows, we find that flows primarily respond to the alpha component obtained with the CP model, but not to the beta component (as would a CAPM investor do).

The remainder of the paper is as follows. Section 2 presents the framework for decomposing hedge fund returns. Section 3 describes the methodology. Section 4 presents the hedge fund dataset and the models. Section 5 contains the empirical analysis, and Section 6 concludes. The appendix provides additional information on the methodology, the data, and the empirical results.

2. Decomposing hedge fund returns under misspecification

2.1. Theoretical framework

2.1.1. Definition of the alpha and beta components

We consider a population of n funds over T periods and denote by $r_{i,t}$ the excess net-of-fee return that fund i ($i = 1, \dots, n$) delivers to investors at time t ($t = 1, \dots, T$). Our objective is to decompose the average fund return $E[r_{i,t}]$ into its alpha and beta components. The definition of each component hinges on its information source. The alpha component ac_i^* is based on private information—it captures the net return that the fund produces by trading on its proprietary information signals. The beta component bc_i^* is based on public information—it captures the return that the fund produces by following mechanical trading strategies. Put differently, bc_i^* represents the return component that sophisticated investors with access to public information can replicate.

To formalize this intuition, we denote the excess return vector of the mechanical strategies by f_t and its average value by $E[f_t] = \lambda$. If the factors f_t are known, we can decompose the average fund return as

$$E[r_{i,t}] = ac_i^* + bc_i^* = \alpha_i^* + b_i^* \lambda, \quad (1)$$

where ac_i^* is given by the fund alpha α_i^* and $bc_i^* = b_i^* \lambda$ corresponds to the average return of a benchmark portfolio with the same betas on

the mechanical strategies as the fund.³ This decomposition allows the number of factors to be large as f_t includes all the alternative strategies followed by hedge funds. In particular, f_t includes the returns of dynamic strategies (or managed portfolios). Therefore, the use of constant betas is not restrictive for capturing factor-timing strategies based on public information (see, e.g., Ferson and Schadt, 1996; Haddad et al., 2020; Kelly and Pruitt, 2013). To elaborate, suppose that hedge funds commonly change their market betas after observing a public signal z_{t-1} that predicts the equity market return $r_{m,t}$. Equation (1) absorbs the time-variation in betas by including the scaled factor $z_{t-1} r_{m,t}$ in the vector f_t .⁴

The magnitude of the beta component depends on the factor premia λ . Consistent with standard asset pricing models, these premia can be a compensation for fundamental sources of risk in the economy. They can also reflect limits to arbitrage when hedge funds bear nonfundamental risk to correct price anomalies caused by behavioral biases (Gromb and Vayanos, 2010). Finally, they can be the outcome of market segmentation when hedge funds require a premium for holding stocks that constrained investors cannot buy (e.g., Merton, 1987). We remain agnostic on the drivers of the premia— bc_i^* simply controls for strategies that investors can replicate using public information.

2.1.2. Remarks about the alpha and beta components

The return decomposition in Equation (1) calls for several comments. First, the alpha component ac_i^* is a measure of performance, not skill. Whereas the two notions are commonly used interchangeably, they differ in important ways (Barras et al., 2022; Berk and van Binsbergen, 2015). Skill determines whether funds are able to create value by exploiting superior information. Performance determines whether investors hold sufficient bargaining power to receive some of this value. Positive performance implies positive skill, but not vice-versa.

Second, Equation (1) ignores the short-term variations in alpha (around its average α_i^*) that could arise from changing economic conditions, industry competition, or aggregate mispricing (e.g., Avramov et al., 2013; Pástor et al., 2015, 2017). As discussed below, measuring α_i^* for hedge funds is a challenging task. Modeling its conditional variation via a proper choice of predictors and functional forms makes the estimation even more difficult (see, e.g., Gagliardini et al., 2020; Bakalli et al., 2023).

Third, Equation (1) does not require that we model the determinants of ac_i^* and bc_i^* across funds. Both components may depend on several fund characteristics such as fees, share restrictions, or leverage.⁵ Specifying these characteristics is useful for identifying funds with high alpha or high exposures to specific strategies (as discussed in Section 5.3.1). However, it is not necessary for estimating the two components ac_i^* and bc_i^* .

2.1.3. The distributions of the alpha and beta components

Hedge funds follow diverse alternative strategies and rely on multiple information signals to create value. It is therefore likely that the alpha and beta components vary across funds. This heterogeneity cannot be captured with a simple average—instead, it requires that we

³ A key requirement is that all elements in f_t are excess returns (i.e., tradable factors with zero prices). Otherwise, $\lambda \neq E[f_t]$, implying that the difference $E[r_{i,t}] - b_i^* E[f_t]$ cannot be interpreted as the alpha α_i^* (see Ferson, 2013).

⁴ This reasoning is also valid if the public signal is observed multiple times between $t-1$ and t . For instance, suppose that we observe fund returns at a monthly frequency and that hedge funds change their market betas using daily signals z_{t-1}^d ($d = 1, 2, \dots$). In this case, we include the scaled factor $\sum_d z_{t-1}^d r_{m,t}^d$ (ignoring compounding for simplicity), where $r_{m,t}^d$ denotes the daily stock return (Patton and Ramadorai, 2013).

⁵ To see how leverage changes the alpha and beta components across funds consider a set of hedge funds with access to the same active strategy whose average net excess return is equal to $\alpha^* + b^* \lambda = ac^* + bc^*$. For each fund i , both ac_i^* and bc_i^* scale up with its leverage ratio π_i : $ac_i^* = \pi_i ac^*$ and $bc_i^* = \pi_i bc^*$.

estimate the entire cross-sectional distributions characterized by their densities ϕ_{ac}^* and ϕ_{bc}^* .

Measuring fund heterogeneity is important for several reasons. The alpha distribution ϕ_{ac}^* determines how many funds deliver positive alphas—a key quantity to test the equilibrium predictions of asset management models. It is also useful for hedge fund investors who only select a handful of funds because of multiple frictions (Bollen et al., 2021). The alpha distribution allows these investors to determine the range of performance outcomes when selecting funds. In other words, ϕ_{ac}^* has a natural Bayesian interpretation as it provides prior information about individual fund alphas (e.g., Jones and Shanken, 2005; Pástor and Stambaugh, 2002).

The distribution ϕ_{bc}^* measures how many funds follow mechanical strategies. It also determines whether funds with the same investment style have similar risk profiles. While this information obviously matters for researchers and investors, it is also relevant for regulators. As discussed by Brown et al. (2010, ch. 12), hedge funds can contribute to systemic risk when they liquidate positions, reduce liquidity provision, or impose losses on counterparties. Contrary to a simple average, ϕ_{bc}^* can identify clusters of funds with strong exposures to similar factors.

2.2. The misspecification of hedge fund models

2.2.1. The prevalence of misspecification

The theoretical decomposition in Equation (1) is important for defining the alpha and beta distributions ϕ_{ac}^* and ϕ_{bc}^* . However, it does not provide guidance for estimating these distributions for two reasons. First, we do not know the identity of the factors in f_t . Second, data limitation and overfitting issues favor models with a parsimonious number of factors.

Choosing appropriate factors is a daunting task as hedge funds invest in many countries and asset classes (Lhabitant, 2007). They take nonlinear option positions that are difficult to capture with a limited set of option factors (Karehnke and de Roan, 2020). Hedge funds also follow multiple dynamic strategies in which the portfolio composition or leverage ratio responds to changing economic conditions (Ang et al., 2011; Avramov et al., 2013; Bollen and Whaley, 2009). Based on this analysis, there is no hope of capturing the average returns of all the alternative hedge fund strategies. As a result, the empirical return decomposition is likely based on a misspecified model—that is, a model that omits some relevant factors in f_t .⁶

Misspecification also arises when measuring how hypothetical investors with different sophistication levels value the alpha and beta components. Whereas the most sophisticated investors only care about ac_i^* , less sophisticated ones also value bc_i^* because it provides exposure to alternative strategies that are difficult to replicate (e.g., Agarwal et al., 2018). To evaluate hedge funds from the viewpoint of a particular investor, we need a model that only includes the factors that she can replicate. This model is therefore misspecified because it only includes a subset of f_t .

2.2.2. The dual impact of misspecification

Misspecification has a dual impact on the estimation of the distributions ϕ_{ac}^* and ϕ_{bc}^* . First, the estimation is imperfect. To see this point, suppose that we use a misspecified model k that only includes the factors $f_{i,t}^k$, but omits the factors $f_{i,t}^k$ (with $f_t = (f_{i,t}^k, f_{i,t}^k)'$). From Equation (1), we have $E[r_{i,t}] = ac_i^* + bc_i^* + bc_{i,O}^*$, where $bc_{i,I}^* = b_{i,I}^{*'} \lambda_I^k$ and $bc_{i,O}^* = b_{i,O}^{*'} \lambda_O^k$ are the beta components due to the included and omitted factors under the true model (i.e., $b_{i,I}^*$ and $b_{i,O}^*$ are the true

fund exposures to $f_{i,t}^k$ and $f_{i,t}^k$). The key question is how the omitted beta component $bc_{i,O}^*$ affects the return decomposition obtained with the misspecified model, which we write as $E[r_{i,t}] = ac_i^k + bc_i^k$.

Regressing the omitted factors on the included factors, we have $f_{i,t}^k = \alpha_O^k + \Psi_{O,I}^k f_{i,t}^k + u_{i,t}^k$ and $\lambda_O^k = \alpha_O^k + \Psi_{O,I}^k \lambda_I^k$, where α_O^k is the vector of factor alphas, $\Psi_{O,I}^k$ is the matrix of slope coefficients, and $u_{i,t}^k$ is the vector of errors. We can split the omitted beta component into two parts: $bc_{i,O}^* = b_{i,O}^{*'} \alpha_O^k + b_{i,O}^{*'} \Psi_{O,I}^k \lambda_I^k$, where the split depends on the correlation between the included and omitted factors (captured by $\Psi_{O,I}^k$). The first part, which arises from the component of $f_{i,t}^k$ that is orthogonal to $f_{i,t}^k$, is absorbed by ac_i^k . The second part, which arises from the component of $f_{i,t}^k$ that is spanned by $f_{i,t}^k$, is absorbed by bc_i^k . As a result, we have:

$$ac_i^k = ac_i^* + b_{i,O}^{*'} \alpha_O^k = \alpha_i^k, \quad (2)$$

$$bc_i^k = bc_i^* - b_{i,O}^{*'} \alpha_O^k = b_{i,I}^{*'} \lambda_I^k, \quad (3)$$

where α_i^k and $b_{i,I}^k = b_{i,I}^{*'} + \Psi_{O,I}^{k'} b_{i,O}^*$ are the coefficients from the regression of the fund return $r_{i,t}$ on the factors $f_{i,t}^k$ included in model k (and a constant). Equations (2)–(3) reveal that ac_i^k and bc_i^k are informative about the true components ac_i^* and bc_i^* . However, this information is polluted—if the fund loads on alternative strategies with positive alphas, $b_{i,O}^{*'} \alpha_O^k$ is positive and the model-implied alpha component is inflated ($ac_i^k > ac_i^*$ and $bc_i^k < bc_i^*$). Misspecification produces an imperfect estimation because we can only infer the pseudo-true distributions ϕ_{ac}^k and ϕ_{bc}^k , but not the true ones ϕ_{ac}^* and ϕ_{bc}^* (Gourieroux et al., 1984; White, 1982).

The second impact of misspecification pertains to estimation noise. When model k is misspecified, the densities ϕ_{ac}^k and ϕ_{bc}^k are estimated with substantial noise. The simplest way to illustrate this point is to focus on the average alpha $M_{1,ac}^k = \int_{-\infty}^{+\infty} x \phi_{ac}^k(x) dx$, where $\phi_{ac}^k(x)$ is the density evaluated at x . To estimate $M_{1,ac}^k$, we run the linear regression under model k for each fund:

$$r_{i,t} = \alpha_i^k + b_{i,I}^{k'} f_{i,t}^k + \varepsilon_{i,t}^k. \quad (4)$$

We then compute the cross-sectional mean $\hat{M}_{1,ac}^k = \frac{1}{n} \sum_i \hat{ac}_i^k = \frac{1}{n} \sum_i \hat{\alpha}_i^k$.

Contrary to the correctly specified case, the error terms $\varepsilon_{i,t}^k$ in Equation (4) are strongly cross-sectionally correlated because they all depend on the common omitted factors $f_{i,t}^k$. Formally, we have $\varepsilon_{i,t}^k = \varepsilon_{i,t}^* + b_{i,O}^{*'} u_{i,t}^k$, where $u_{i,t}^k$ is the error vector of $f_{i,t}^k$. Even if the hedge fund population size n is large, the information is limited because the values of $\hat{ac}_i^k$ ($i = 1, \dots, n$) are all impacted by the random realizations of the omitted component $u_{i,t}^k$. It implies that $\hat{M}_{1,ac}^k$ is estimated with substantial noise, as discussed in detail in Section 3.

2.3. A new approach for addressing misspecification

2.3.1. Overview of the approach

We develop a novel approach to estimate the entire alpha and beta distributions for any model. A key feature of our approach is to explicitly account for the dual impact of misspecification. First, it allows for an examination of multiple models to address the imperfect separation between alpha and beta. Given the diversity in hedge fund strategies, several models can reasonably be used for decomposing returns. By designing formal comparison tests, we identify models less prone to misspecification and more able to capture the true distributions ϕ_{ac}^* and ϕ_{bc}^* .⁷

⁶ One could use holdings-based benchmarks to avoid specifying the factors (e.g., Grinblatt and Titman, 1993; Lo, 2008). However, these measures are still subject to misspecification when the fund exhibits time-varying betas (Ferson and Khang, 2002). Another issue is that hedge funds generally do not disclose their portfolio weights.

⁷ Another benefit of this comparison analysis is to measure how hypothetical investors with different sophistication levels value hedge fund investments—an issue that we examine empirically in Section 5.5.

Second, our approach comes with a full-fledged inferential theory to incorporate the estimation noise caused by misspecification. We derive the asymptotic properties of the estimated distributions in a setting that accounts for the large population of hedge funds observed in the data (i.e., we let n to grow large). Using these results, we (i) conduct proper statistical inference on the distributions for each model, and (ii) obtain valid tests for comparing models.

Our approach departs from previous studies that estimate the alpha distribution (e.g., Chen et al., 2017; Harvey and Liu, 2018). First, it does not assume that the model is correctly specified—an important feature given the difficulty in modeling hedge fund returns. Second, it provides a framework for estimating both the alpha and beta distributions. Third, it is flexible because it imposes no restrictions on the shape of these distributions. Fourth, it is simple and fast as it does not rely on sophisticated and computer-intensive Gibbs sampling and expectation maximization methods. Last but not least, it allows us to conduct statistical inference and formal testing guided by econometric theory.

2.3.2. Comparing models using the CAPM

The comparison of models ultimately hinges on their ability to capture alternative strategies. These strategies, which are weakly correlated with the equity market, represent a defining feature of hedge funds.⁸ Superior models capture the positive premia earned on alternative strategies and thus deliver alpha components close to zero. In contrast, highly misspecified models are unable to do so and produce larger alphas. To distinguish between these models, we use the CAPM as a reference—that is, we formally compare the alphas under any proposed model k and the CAPM.

The CAPM is a natural reference for two reasons. First, it controls for the equity market risk taken by hedge funds—an opener evaluation practice among academics and practitioners. Second, it is the simplest and thus the least equipped model for capturing alternative strategies. To formalize this point, we refer to the CAPM as model 0 and denote by $f_{i,t}^0$ the market factor and by $f_{O,t}^0$ the set of alternative strategies. If $f_{i,t}^0$ and $f_{O,t}^0$ are uncorrelated ($\Psi_{O,I}^0 = 0$), we have $\alpha_O^0 = \lambda_O^0$. As a result, the CAPM alpha entirely absorbs the beta component $bc_{i,O}^*$ due to the alternative strategies:

$$ac_i^0 = ac_i^* + b_{i,O}^{*f} \alpha_O^0 = ac_i^* + b_{i,O}^{*f} \lambda_O^0 = ac_i^* + bc_{i,O}^*. \quad (5)$$

Building on this insight, our comparison analysis delivers a strong message. If the alpha distributions ϕ_{ac}^k and ϕ_{ac}^0 are identical, it implies that model k is as misspecified as the CAPM. In other words, model k is unable to capture any of the alternative strategies followed by hedge funds.⁹

2.3.3. A simple illustrative example

Before presenting our approach in more detail, we briefly illustrate its usefulness using a simple example. We assume that the correct hedge fund model includes the market and three uncorrelated alternative factors: $E[r_{i,t}] = \alpha_i^* + b_{i,m}^* \lambda_m + b_{i,1}^* \lambda_1 + b_{i,2}^* \lambda_2 + b_{i,3}^* \lambda_3$, where λ_m is the equity premium, and λ_j denotes the premium of each alternative factor j ($j = 1, 2, 3$), which we set equal to λ_m . For each fund, the alpha component α_i^* is drawn from a normal $N(\mu_\alpha^*, \sigma_\alpha^{*2})$, $b_{i,m}^*$ from a normal $N(\mu_b^*, \sigma_b^{*2})$, and $b_{i,j}^*$ from a normal $N(\mu_{b_j}^*, \sigma_{b_j}^{*2})$, where $\mu_{b_j}^*$ is positive to capture the view that hedge funds load on alternative strategies (each

draw is mutually independent). We further assume that the first factor is more important by setting $\mu_{b_1}^* = \mu_b^*$ and $\mu_{b_2}^* = \mu_{b_3}^* = \mu_b^*/3$.

We consider a misspecified hedge fund (HF) model (model 1) that includes the market and two out of the three alternative factors (f_1 and f_2). We compare this model with the CAPM (model 0), which only includes the market. For each model, we assume that we observe the fund alpha and beta components without errors, leaving aside estimation noise for now. Given the above assumptions, the alpha and beta densities are normal. Under the CAPM, we have

$$ac_i^0 \sim N(\mu_\alpha^* + (\mu_{b_1}^* + \mu_{b_2}^* + \mu_{b_3}^*)\lambda, \sigma_\alpha^{*2} + 3\sigma_b^{*2}\lambda^2), \quad (6)$$

$$bc_i^0 \sim N(\mu_b^* \lambda, \sigma_b^{*2}\lambda^2), \quad (7)$$

while the HF model produces the following densities:

$$ac_i^1 \sim N(\mu_\alpha^* + \mu_{b_3}^* \lambda, \sigma_\alpha^{*2} + \sigma_b^{*2}\lambda^2), \quad (8)$$

$$bc_i^1 \sim N((\mu_b^* + \mu_{b_1}^* + \mu_{b_2}^*)\lambda, 3\sigma_b^{*2}\lambda^2). \quad (9)$$

To begin, we plot in Panel A of Fig. 1 the two alpha densities using the following parameter values: $\mu_\alpha^* = 0\%$, $\sigma_\alpha^* = 1.4\%$, $\lambda = 7.5\%$, $\mu_b^* = 0.3$, and $\sigma_b^* = 0.4$.¹⁰ The comparison of ϕ_{ac}^0 and ϕ_{ac}^1 reveals that the magnitude of the alpha components decreases substantially under the HF model (ϕ_{ac}^1 moves to the left towards zero). This difference arises because the HF model captures two alternative strategies (out of three) and thus produces a sharper identification of ac_i^* and bc_i^* . The implications for performance evaluation are economically important. Under the CAPM, the average alpha reaches 3.8% per year. Under the HF model, the average drops to 0.8% per year and moves closer to the true average μ_α^* equal to zero.

In addition to reducing the magnitude of the alphas, the HF model also produces a lower dispersion as it absorbs the variation due to factors 1 and 2 (the term $2\sigma_b^{*2}\lambda^2$). However, this result depends on the specific assumptions in our simple example—in particular, we assume that funds choose similar factor exposures regardless of their true alphas (i.e., we set $\text{corr}[\alpha_i^*, b_{i,j}^*] = 0$). If we relax this assumption, a lower average does not necessarily come with a lower dispersion. In the empirical analysis, we simply let the data speak—our approach is nonparametric and thus does not require any assumptions on the joint distributions of α_i^* and $b_{i,j}^*$.

Next, we repeat the analysis for the two beta densities ϕ_{bc}^0 and ϕ_{bc}^1 . Consistent with intuition, Panel B is the mirror image of Panel A as the magnitude of the beta components rises under the HF model (ϕ_{bc}^1 moves to the right away from zero). We can go one step further and decompose bc_i^1 as $bc_{i,m}^1 + bc_{i,1}^1 + bc_{i,2}^1$, where $bc_{i,m}^1$ is the beta component due to the market, and $bc_{i,1}^1$, $bc_{i,2}^1$ are the beta components due to the two additional factors in the HF model. We can then determine the economic importance of these factors using their densities $\phi_{bc_1}^1$ and $\phi_{bc_2}^1$, which are both normal: $bc_{i,1}^1 \sim N(\mu_{b_1}^* \lambda, \sigma_b^{*2}\lambda^2)$ and $bc_{i,2}^1 \sim N(\mu_{b_2}^* \lambda, \sigma_b^{*2}\lambda^2)$. This analysis identifies factor 1 as the most relevant—its contribution to hedge fund returns equals 2.3% per year on average and is positive for 77% of the funds. In contrast, these values are only equal to 0.8% and 60% for factor 2.

3. Methodology

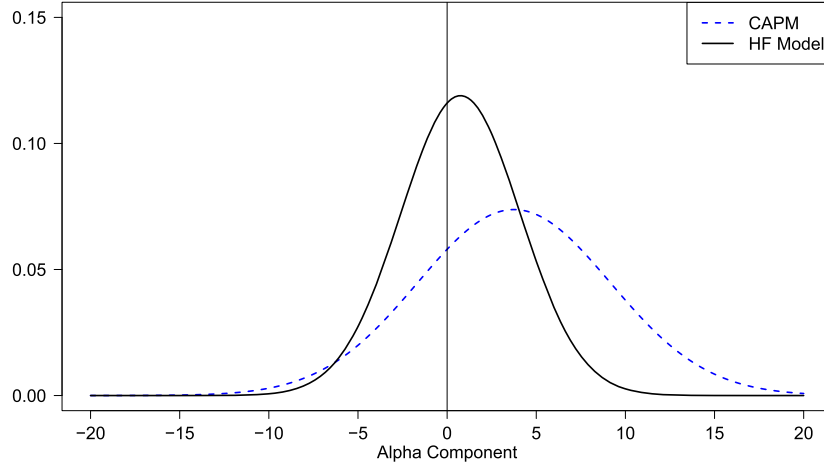
3.1. Estimation procedure

We consider a total of K models indexed by k ($k = 0, \dots, K - 1$), where model 0 denotes the CAPM. Each model k is allowed to be mis-

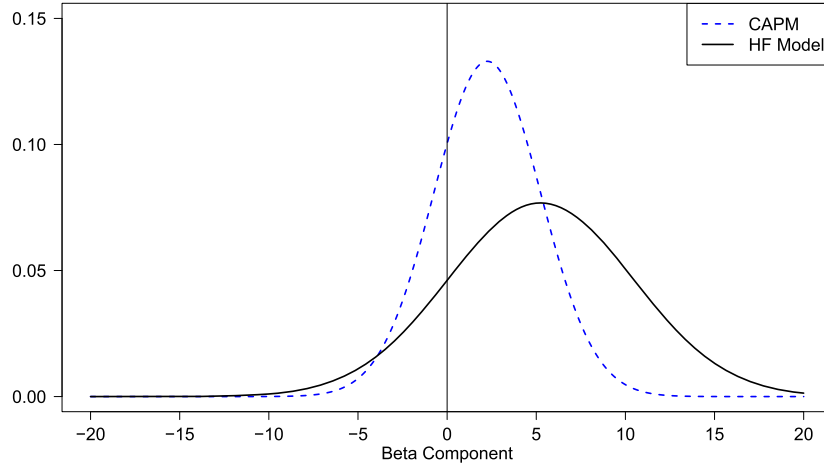
⁸ For instance, Carhart et al. (2014) define alternative (exotic) betas as exposures to risk factors that are uncorrelated with global equity markets and have positive expected returns. They also note that “it is fair to characterize hedge funds as providing considerable exposure to equity market risk and to alternative risk premiums.”

⁹ Whereas we use the CAPM for the baseline analysis, our comparison tests can be applied to any pair of models. We can therefore replace the CAPM with any model commonly used for performance evaluation (see Section 5.1.1).

¹⁰ We express μ_α^* , σ_α^* , and λ in percent per year. For simplicity, we set μ_α^* equal to zero and calibrate σ_α^* using the value reported by Barras et al. (2022, Table VI). We further set λ equal to the average market return and μ_b , σ_b equal to the cross-sectional mean and volatility of the market betas in our sample of hedge funds.



(a) Distribution of the Alpha Components



(b) Distribution of the Beta Components

Fig. 1. Distributions of the Alpha and Beta Components – A Simple Example.

This figure compares the average return decomposition obtained with a candidate hedge fund (HF) model and the CAPM. Panel A plots the cross-sectional distributions of the alpha components (annualized) under both models. In this simple example, the average fund returns are explained by four factors (the market and three alternative factors 1, 2, and 3 with similar premia), and hedge funds load more aggressively on factor 1 than on factors 2 and 3. Whereas the CAPM omits factors 1, 2, and 3, the HF model only omits factor 3. Panel B repeats the analysis for the beta components.

specified as it includes the factors $f_{I,t}^k$ but omits the factors $f_{O,t}^k$. We are interested in (i) the density ϕ_{ac}^k of the alpha component, (ii) the density ϕ_{bc}^k of the beta component, and (iii) the density $\phi_{bc,j}^k$ of the beta component due to each factor j included in model k . We summarize the shape of each of these densities with the following characteristics: (i) the cross-sectional mean and standard deviation, denoted by M_1^k and M_2^k , (ii) the proportion of funds with a return component below a given value a , denoted by $P^k(a)$, and (iii) the quantile at a given percentile level u , denoted by $Q^k(u) = (P^k)^{-1}(u)$.

To estimate the distribution characteristics of each density, we need to estimate the fund components $ac_i^k = \alpha_i^k$, $bc_i^k = b_{i,I}^k \lambda_{I,j}^k$, and $bc_{i,j}^k = b_{i,I,j}^k \lambda_{I,j}^k$, where $b_{i,I,j}^k$ and $\lambda_{I,j}^k$ is the beta and premium associated with factor j in model k . For each fund, we compute these values by running the time-series regression in Equation (4). We interpret this regression as a random coefficient model (e.g., Hsiao, 2003) in which α_i^k and $b_{i,I}^k$ are not fixed parameters, but random realizations from a continuum of

funds in order to invoke cross-sectional limits.¹¹ We also assume that at least one omitted factor is strong in the sense that it has a pervasive impact on the cross-section of fund returns.¹² This mild assumption, which delivers a well-defined convergence rate for the distribution characteristics, is supported by the diagnostic criterion of Gagliardini et al. (2019) (see Section 4.3). The least-square estimate $\hat{\gamma}_i^k = (\hat{\alpha}_i^k, \hat{b}_{i,I}^k)'$ is given by

$$\hat{\gamma}_i^k = (\hat{Q}_{x,i}^k)^{-1} \frac{1}{T_i} \sum_t I_{i,t} x_{i,t}^k r_{i,t}, \quad (10)$$

¹¹ Barras et al. (2022) and Gagliardini et al. (2016) use the same sampling scheme to measure mutual fund performance and test the arbitrage pricing theory in a large cross-section of assets.

¹² More formally, an omitted factor is strong if the largest eigenvalue of the residual covariance matrix of hedge fund returns does not vanish as the population size n grows large. In contrast, a factor is weak if its loading vanishes at a rate $1/\sqrt{T}$ (Gagliardini et al., 2019). Our formulation allows for weak factors in both $f_{I,t}^k$ and $f_{O,t}^k$.

where $I_{i,t}$ is an indicator variable equal to one if $r_{i,t}$ is observable, $T_i = \sum_t I_{i,t}$, $x_i^k = (1, f_{i,t}^k)'$, and $\hat{Q}_{x,i}^k = \frac{1}{T_i} \sum_t I_{i,t} x_i^k x_i^{k'}$. We then compute the alpha and beta components for each fund as

$$\hat{a}c_i^k = \hat{\alpha}_i^k, \quad (11)$$

$$\hat{b}c_i^k = \hat{\mu}_i - \hat{\alpha}_i^k, \quad (12)$$

$$\hat{b}c_{i,j}^k = \hat{b}_{i,j}^k \hat{\lambda}_{i,j}^k, \quad (13)$$

where $\hat{b}c_i^k$ is the empirical counterpart of $bc_i^k = E[r_{it}] - ac_i^k$, $\hat{\mu}_i = \frac{1}{T_i} \sum_t I_{i,t} r_{i,t}$ is the average fund return, and $\hat{\lambda}_{i,j}^k = \frac{1}{T_i} \sum_t I_{i,t} f_{i,t}^k f_{i,t}^{k'}$ is the empirical average of $f_{i,t}^k f_{i,t}^{k'}$.

Next, we account for the unbalanced nature of the hedge fund sample. Following Barras et al. (2022) and Gagliardini et al. (2016), we introduce a formal selection rule $\mathbf{1}_i^x$ equal to one if the following conditions are met: $\mathbf{1}_i^x = \mathbf{1} \{ \tau_{i,T} \leq \chi_{1,T}, CN_i \leq \chi_{2,T} \}$, where $\tau_{i,T} = T/T_i$, $CN_i = \sqrt{\text{eig}_{\max}(\hat{Q}_{x,i}^k) / \text{eig}_{\min}(\hat{Q}_{x,i}^k)}$ is the condition number of $\hat{Q}_{x,i}^k$, and $\chi_{1,T}$, $\chi_{2,T}$ denote the two threshold values. The first condition $\tau_{i,T} \leq \chi_{1,T}$ excludes funds for which the sample size is too small. The second condition $CN_i \leq \chi_{2,T}$ excludes funds for which the time-series regression is subject to multicollinearity problems (e.g., Belsley et al., 2004). Both thresholds $\chi_{1,T}$ and $\chi_{2,T}$ increase with the sample size T —with more return observations, we estimate the fund coefficients with greater accuracy, which allows for a less stringent selection rule. Applying this selection rule, we work with a population size equal to $n_x = \sum_{i=1}^n \mathbf{1}_i^x$.

The final step is to compute the distribution characteristics using the vector of estimated components. We compute the mean, standard deviation, proportion, and quantile of the alpha distribution as $\hat{M}_1^k = \frac{1}{n_x} \sum_i \mathbf{1}_i^x \hat{a}c_i^k$, $\hat{M}_2^k = \left(\frac{1}{n_x} \sum_i \mathbf{1}_i^x (\hat{a}c_i^k)^2 - \left(\frac{1}{n_x} \sum_i \mathbf{1}_i^x \hat{a}c_i^k \right)^2 \right)^{1/2}$, $\hat{P}^k(a) = \frac{1}{n_x} \sum_i \mathbf{1}_i^x \mathbf{1} \{ \hat{a}c_i^k \leq a \}$, and $\hat{Q}^k(u) = (\hat{P}^k)^{-1}(u)$. To obtain the characteristics of the distributions of the beta components, we use the same formulas after replacing $\hat{a}c_i^k$ with $\hat{b}c_i^k$ or $\hat{b}c_{i,j}^k$.

3.2. Properties of the distribution characteristics

We begin our theoretical analysis by examining the properties of the distribution characteristics $\hat{M}_1^k$, $\hat{M}_2^k$, $\hat{P}^k(a)$, and $\hat{Q}^k(u)$. The following proposition derives the asymptotic distribution of the estimated characteristics of the alpha distribution ϕ_{ac}^k as the numbers of funds n and observations T grow large. To capture the large cross-sectional dimension of the hedge fund population observed in the data, we require that n is larger than T .

Proposition 1. As $n, T \rightarrow \infty$, such that $T/n \rightarrow 0$, we obtain the following properties for the estimated characteristics of ϕ_{ac}^k under the misspecified model k :

$$\sqrt{T} (\hat{M}_s^k - M_s^k) \rightarrow_d N(0, V[M_s^k]), \quad (14)$$

$$\sqrt{T} (\hat{P}^k(a) - P^k(a)) \rightarrow_d N(0, V[P^k(a)]), \quad (15)$$

$$\sqrt{T} (\hat{Q}^k(u) - Q^k(u)) \rightarrow_d N(0, V[Q^k(u)]), \quad (16)$$

where $s \in \{1, 2\}$ and $\rightarrow_d$ denotes convergence in distribution. The variance terms are given by

$$V[M_s^k] = \left(\eta_{M_s}^{k'} \otimes E_1'(Q_x^k)^{-1} \right) \Omega_{ux}^k \left(\eta_{M_s}^k \otimes (Q_x^k)^{-1} E_1 \right), \quad (17)$$

$$V[P^k(a)] = \left(\eta_{P(a)}^{k'} \otimes E_1'(Q_x^k)^{-1} \right) \Omega_{ux}^k \left(\eta_{P(a)}^k \otimes (Q_x^k)^{-1} E_1 \right), \quad (18)$$

$$V[Q^k(u)] = V[P^k(Q^k(u))]/\phi_{ac}^k(Q^k(u))^2, \quad (19)$$

where $\eta_{M_s}^k = E \left[\left(\frac{\partial M_s^k}{\partial E[g_i^k]} \right)' \frac{\partial g_i^k}{\partial ac_i^k} b_{i,O}^* \right]$, $E[g_i^k]$ is the vector of uncentered moments with $g_i^k = (ac_i^k, (ac_i^k)^2)'$, $\otimes$ denotes the Kronecker product, $Q_x^k = E[x_i^k x_i^{k'}]$, $\Omega_{ux}^k = \lim_{T \rightarrow \infty} V \left[\frac{1}{\sqrt{T}} \sum_t u_{O,t}^k \otimes x_t^k \right]$, $\eta_{P(a)}^k = E[b_{i,O}^* | ac_i^k = a] \phi_{ac}^k(a)$, $b_{i,O}^*$ and $u_{O,t}^k$ denote the vectors of betas and residuals associated with the omitted factors $f_{O,t}^k$, and $\phi_{ac}^k(a)$ is the probability density evaluated at a .

Proof. See the appendix.

To save space, we refer the reader to the appendix for the theoretical analysis of the distributions ϕ_{bc}^k and $\phi_{bc,j}^k$ whose estimated characteristics have the same properties as in Proposition 1. These results allow for formal tests on the shape of the alpha or beta distribution. Denoting the generic estimated characteristic by $\hat{C}^k \in \{\hat{M}_s^k, \hat{P}^k(a), \hat{Q}^k(u)\}$, we can test the null hypothesis:

$$H_0 : C^k = v, \quad (20)$$

where v is a given scalar. For instance, we can test whether the proportion of positive-alpha funds equals 50% ($v = 0.5$), or whether the average beta component due to any factor j is null ($v = 0$).

Proposition 1 reveals two important insights about inference under misspecification. First, the estimated characteristics converge towards their respective parameter values. Asymptotically, we are able to estimate the alpha distribution ϕ_{ac}^k under model k without bias, even though we use as inputs noisy versions of the fund components (i.e., we use $\hat{a}c_i^k$ instead of ac_i^k). Under misspecification, Proposition 1 therefore provides a theoretical justification for the common practice of reporting cross-sectional summary statistics (e.g., boxplots) based on estimated coefficients without any bias adjustment.¹³ Second, the variance of the estimators is large because the convergence rate is equal to $1/\sqrt{T}$ (and not $1/\sqrt{n}$)—a result that formalizes our previous point that misspecification amplifies estimation noise. This result is a priori surprising because the characteristics are all computed as cross-sectional averages (we sum across n funds, not across T periods).

None of these properties hold when the model is correctly specified—a setting examined in detail by Barras et al. (2022). In this case, the estimated distribution characteristics must be adjusted for the error-in-variable (EIV) bias that arises because we use as inputs noisy versions of the fund coefficients. In addition, the characteristics are estimated with greater precision because the convergence rate is equal to $1/\sqrt{n}$ (instead of $1/\sqrt{T}$).

The strong impact of misspecification on inference stems from the properties of the fund error terms. The estimation error on the alpha component $\hat{a}c_i^k$ involves the term $\bar{\varepsilon}_i^k = \bar{\varepsilon}_i^* + b_{i,O}^{*'} \bar{u}_O^k$, where $\bar{\varepsilon}_i^k$, $\bar{\varepsilon}_i^*$, and $\bar{u}_O^k$ denote the time-series averages of $\varepsilon_{i,t}^k$, $\varepsilon_{i,t}^*$, and $u_{O,t}^k$. The average $\bar{\varepsilon}_i^*$ obtained with the correct model is weakly correlated across funds, which implies that its impact on the estimated characteristics vanishes with the population size n . It is not the case for the average error term $\bar{u}_O^k$ of the omitted factors because it affects all funds simultaneously. This term is noisy because it converges to zero at the rate equal to $1/\sqrt{T}$. As a result, the noise contained in $\bar{u}_O^k$ (i) slows down the convergence rate of the estimated characteristics from $1/\sqrt{n}$ to $1/\sqrt{T}$, and (ii) dominates in magnitude the EIV bias, which makes any bias adjustment unnecessary.¹⁴

¹³ For instance, Almeida et al. (2020), Capocci and Hübner (2004), and Kosowski et al. (2007) follow this practice when reporting the characteristics of the distribution of hedge fund alphas.

¹⁴ Our Monte Carlo simulations calibrated on our sample of funds confirm these results in finite samples (see the appendix). When the model is misspecified, the mean squared error (MSE) of each characteristic estimator (i)

3.3. Comparison tests between models

We now turn to the comparison tests based on the alpha distributions. We compare the distributions ϕ_{ac}^k and ϕ_{ac}^l between two models k and l ($k, l = 0, \dots, K - 1$). Our comparison framework is highly flexible because we can apply it to both nested and non-nested models (models are nested if one is included in the other). When we set $l = 0$, the comparison is made with respect to the CAPM. We compute the differences in distribution characteristics between the two models as $\Delta \hat{M}_1 = \hat{M}_1^k - \hat{M}_1^l$, $\Delta \hat{M}_2 = \hat{M}_2^k - \hat{M}_2^l$, $\Delta \hat{P}(a) = \hat{P}^k(a) - \hat{P}^l(a)$, and $\Delta \hat{Q}(u) = \hat{Q}^k(u) - \hat{Q}^l(u)$. Proposition 2 derives the asymptotic distribution of each estimated difference as the numbers of funds n and observations T grow large.

Proposition 2. *As $n, T \rightarrow \infty$ such that $T/n \rightarrow 0$, we obtain the following properties for the differences in the estimated characteristics of ϕ_{ac}^k and ϕ_{ac}^l under the misspecified models k and l :*

$$\sqrt{T} (\Delta \hat{M}_s - \Delta M_s) \rightarrow_d N(0, V[\Delta M_s]), \quad (21)$$

$$\sqrt{T} (\Delta \hat{P}(a) - \Delta P(a)) \rightarrow_d N(0, V[\Delta P(a)]), \quad (22)$$

$$\sqrt{T} (\Delta \hat{Q}(u) - \Delta Q(u)) \rightarrow_d N(0, V[\Delta Q(u)]), \quad (23)$$

where $s \in \{1, 2\}$. The variance of the characteristic differences are equal to

$$V[\Delta M_s] = V[M_s^k] + V[M_s^l] - 2Cov[M_s^k, M_s^l], \quad (24)$$

$$V[\Delta P(a)] = V[P^k(a)] + V[P^l(a)] - 2Cov[P^k(a), P^l(a)], \quad (25)$$

$$V[\Delta Q(u)] = V[Q^k(u)] + V[Q^l(u)] - 2Cov[Q^k(u), Q^l(u)], \quad (26)$$

where $V[M_s^k]$, $V[M_s^l]$, $V[P^k(a)]$, $V[P^l(a)]$, $V[Q^k(u)]$, and $V[Q^l(u)]$ are obtained from Proposition 1. The covariance terms are given by

$$Cov[M_s^k, M_s^l] = \left(\eta_{M_s}^{kl} \otimes E_1'(Q_x^k)^{-1} \right) \Omega_{ux}^{kl} \left(\eta_{M_s}^l \otimes (Q_x^l)^{-1} E_1 \right), \quad (27)$$

$$Cov[P^k(a), P^l(a)] = \left(\eta_{P(a)}^{kl} \otimes E_1'(Q_x^k)^{-1} \right) \Omega_{ux}^{kl} \left(\eta_{P(a)}^l \otimes (Q_x^l)^{-1} E_1 \right), \quad (28)$$

$$Cov[Q^k(u), Q^l(u)] = \frac{Cov[P^k(Q^k(u)), P^l(Q^l(u))]}{\phi_{ac}^k(Q^k(u))\phi_{ac}^l(Q^l(u))}, \quad (29)$$

$$\text{where } \Omega_{ux}^{kl} = \lim_{T \rightarrow \infty} Cov \left[\frac{1}{\sqrt{T}} \sum_t u_{O,t}^k \otimes x_t^k, \frac{1}{\sqrt{T}} \sum_t u_{O,t}^l \otimes x_t^l \right].$$

Proof. See the appendix.

The results in Proposition 2 provide simple comparison tests for each pair of models k and l . We denote the generic estimated characteristic difference by $\Delta \hat{C} \in \{\Delta \hat{M}_s, \Delta \hat{P}(a), \Delta \hat{Q}(u)\}$. We can then test the null hypothesis that each characteristic difference equals zero:

$$H_0 : \Delta C = 0. \quad (30)$$

Misspecification arises naturally in pairwise comparisons because competing models cannot be all correct. Therefore, the impact of misspecification discussed in Proposition 1 carries over to model comparisons. Proposition 2 shows that the comparison tests inherit the high estimation noise caused by misspecification as each characteristic difference converges at a slow rate equal to $1/\sqrt{T}$. In other words, the bar for detecting significant differences between models is considerably higher when the tests are properly adjusted for misspecification.

Our approach departs from previous studies (e.g., Kan and Robotti, 2009; Kan et al., 2013) which derive comparison tests for misspecified models under a fixed number of assets n (single asymptotics with n fixed

and $T \rightarrow \infty$). These tests require the inversion of the entire covariance matrix of returns—an operation that cannot be performed with thousands of hedge funds, but only hundreds of observations. To address this issue, we derive comparison tests in which n grows large (double asymptotics with n and $T \rightarrow \infty$). Another important difference is that we do not focus on a single aggregate measure of asset alphas (e.g., Hansen-Jagannathan distance), but instead focus on the disaggregated distribution to capture fund heterogeneity.

3.4. Estimation of the asymptotic variance terms

Conducting inference requires consistent estimators of the variance terms in Propositions 1 and 2. For each distribution characteristic, the variance depends on the error term $u_{O,t}^k$ and betas $b_{i,O}^*$ associated with the omitted factors. For instance, the estimated average $\hat{M}_1^k$ is more volatile when the variance of the factor residuals $V[u_{O,t}^k]$ and the magnitude of the average betas $E[b_{i,O}^*]$ increase. Because $u_{O,t}^k$ and $b_{i,O}^*$ are not observable, estimating V is not trivial.

To address this issue, we derive a consistent variance estimator based on the observed fund residuals of each model $\hat{\varepsilon}_{i,t}^k = r_{i,t} - x_{i,t}^{k'} \hat{\gamma}_i^k$. The estimators of the asymptotic variances of $\sqrt{T}(\hat{C}^k - C^k)$ and $\sqrt{T}(\Delta \hat{C} - \Delta C)$ are given by

$$\hat{V}[\hat{C}^k] = \frac{1}{n^2 T} \sum_i \sum_j \sum_t \mathbf{1}_i^x \tau_{i,t} I_{i,t} \mathbf{1}_j^x \tau_{j,t} I_{j,t} \hat{a}_{i,t}(\hat{C}^k) \hat{a}_{j,t}(\hat{C}^k), \quad (31)$$

$$\hat{V}[\Delta \hat{C}] = \frac{1}{n^2 T} \sum_i \sum_j \sum_t \mathbf{1}_i^x \tau_{i,t} I_{i,t} \mathbf{1}_j^x \tau_{j,t} I_{j,t} \hat{a}_{i,t}(\Delta \hat{C}) \hat{a}_{j,t}(\Delta \hat{C}), \quad (32)$$

where the terms $\hat{a}_{i,t}(\hat{C}^k)$ and $\hat{a}_{i,t}(\Delta \hat{C})$ are functions of $\hat{C}^k$ and $\Delta \hat{C}$ (see the appendix). The following proposition shows that $\hat{V}[\hat{C}^k]$ and $\hat{V}[\Delta \hat{C}]$ are consistent variance estimators as the numbers of funds n and observations T grow large.

Proposition 3. *As $n, T \rightarrow \infty$ such that $T/n \rightarrow 0$, we have*

$$\hat{V}[\hat{C}^k] \rightarrow_p V[\hat{C}^k], \quad (33)$$

$$\hat{V}[\Delta \hat{C}] \rightarrow_p V[\Delta \hat{C}], \quad (34)$$

where $\rightarrow_p$ denotes convergence in probability.

Proof. See the appendix.

4. Data and model construction

4.1. Hedge fund dataset

We collect the monthly net-of-fee returns of hedge funds between January 1994 and December 2020. We combine four databases (BarclayHedge, HFR, Morningstar, TASS) to mitigate the selection bias that arises from the voluntary nature of information disclosure by hedge funds. We remove the first 12 months of data for each fund to control for backfill bias. The appendix provides more detail on the construction of the dataset, which largely follows Joenväärä et al. (2021). For comparison purposes, we also collect the monthly net-of-fee returns of actively managed U.S. equity funds over the same period following Barras et al. (2022).

Table 1 reports summary statistics for the equal-weighted portfolio of all hedge funds in our sample, as well as three investment styles: (i) equity funds (long-short and market neutral), which rely on discretionary or quantitative analysis to detect mispriced stocks, (ii) macro funds (global macro and managed futures), which take directional bets across asset classes using broad economic and financial indicators, and (iii) arbitrage funds (relative value and event driven), which exploit various sources of mispricing primarily in the debt market. Overall, the

is primarily driven by the variance (and not by the finite-sample bias), and (ii) decreases with T , but not with n .

Table 1
Summary Statistics for the Equal-Weighted Portfolio of Hedge Funds.

This table provides summary statistics for the equal-weighted portfolio of all existing funds at the start of each month in the entire population and three investment categories: (i) equity funds (long-short, market neutral), (ii) macro funds (global macro, managed futures), and (iii) arbitrage funds (relative value, event driven). We report the annualized mean and standard deviation, skewness, kurtosis, and quantiles at 10% and 90% of the portfolio excess return. The statistics are computed using monthly data between January 1994 and December 2020.

	Moments				Quantiles	
	Mean	Std Dev.	Skewness	Kurtosis	10%	90%
All Funds	5.45	5.49	-0.31	4.51	-1.58	2.34
Equity	6.73	8.70	-0.47	5.32	-2.56	3.30
Long-Short	7.24	9.68	-0.44	5.31	-2.90	3.68
Market Neutral	3.07	2.72	-0.58	5.75	-0.66	1.13
Macro	4.40	6.12	0.40	3.57	-1.68	2.59
Global Macro	5.05	5.79	0.24	3.30	-1.66	2.50
Managed Futures	4.09	6.73	0.47	3.74	-1.79	2.92
Arbitrage	5.40	4.96	-2.16	14.57	-0.97	1.83
Relative Value	4.90	4.47	-2.46	18.03	-0.78	1.60
Event Driven	6.08	6.11	-1.74	11.79	-1.30	2.35

results are similar to those reported by Getmansky et al. (2015) between 1996 and 2014.

To obtain reliable estimates of $ac_{i,j}^k$, $bc_{i,j}^k$, and $bc_{i,j}^k$ in the unbalanced panel of hedge funds, we apply the selection rule of Section 3.1. Taking the same thresholds as Barras et al. (2022), we set the minimum number of return observations equal to 60 and the minimum condition number for each model equal to 15. Empirically, the second threshold plays no role, which implies that the sample of selected funds is identical across all models ($n_x = 5,231$ funds). To address the concern that the results depend on the construction of the hedge fund database, we consider alternative filters in the appendix.¹⁵ Consistent with intuition, we find that these changes have the same impact across models. As a result, they leave the model comparison unchanged.

4.2. Hedge fund models

We apply our methodology to nine models examined in previous work. These models only include tradable factors, which allows us to use the fund-by-fund regression in Equation (4) to compute the alpha and beta components. Our selection focuses on omnibus models that aim at explaining the return of any given hedge fund. We therefore do not include models whose factors vary with each individual fund (e.g., Bollen and Whaley, 2009; O'Doherty et al., 2016).

In addition to the CAPM, we consider four standard models. The chosen set is by no means exhaustive but provides a good representation of the models commonly used for performance evaluation. We select the Carhart (1997) model and the Five-Factor model of Fama and French (2015), which includes the market, size, value, momentum, profitability, and investment factors. We examine the well-known model of Fung and Hsieh (2004), which includes two equity factors (market and size), two bond factors (term and default), and three option straddles (bond, commodity, and currency). Finally, we consider the model of Asness et al. (2013), which adds to the CAPM two global value and momentum factors across international asset classes.

Next, we consider the models of Kozak et al. (2020) based on machine learning. These authors apply lasso and ridge penalizations to

form models with the highest out-of-sample ability to explain the average returns of 50 characteristic-based equity portfolios.¹⁶ We use the two models presented in their Table 4, which impose sparsity by selecting five factors only. The first one is formed with the portfolios themselves, while the second one is formed with their principal components. We also add the market to both models for consistency with the other models.

Finally, we consider two models that include five factors (referred to as additional factors): illiquidity, betting against beta (BAB), variance (short position), carry, and time-series (TS) momentum. These factors are based on economic intuition and plausibly capture several strategies followed by hedge funds.¹⁷ We consider the model of Joenväärä et al. (2021, JKKT), which extends the Carhart (1997) model by including the illiquidity, BAB, and TS momentum factors. Building on the work of Carhart et al. (2014) and Pedersen (2015), we examine another model (CP) which replaces the five non-equity factors of Fung and Hsieh (2004) (bond factors and straddles) with the five additional factors. Measuring the trading costs of these factors is difficult because it requires timely price information on a wide range of markets. Given this uncertainty, we conduct our baseline analysis using the original factor returns.

Table 2 reports summary statistics for the excess returns of the factors.¹⁸ We find that all but one factor (bond term) deliver positive premia. Therefore, hedge funds boost their average returns when they increase their factor betas. Unreported results also show that the factors capture distinct strategies—only 11 pairwise correlations out of 153 are

¹⁵ We change the fund selection rule by imposing 36 or 84 minimum return observations. We also apply the more stringent backfill bias correction of Joenväärä et al. (2021), which eliminates all the observations before the fund listing date. Finally, we use the five filters proposed by Straumann (2009) to remove errors in reported fund returns.

¹⁶ We thank Serhiy Kozak for providing us with the data and code. In principle, we could expand the set of candidate factors using machine learning techniques. However, these techniques require a considerably large number of return observations (Gu et al., 2020), which is not the case for hedge fund datasets.

¹⁷ The illiquidity factor of Pástor and Stambaugh (2003) captures marketwide changes in market liquidity. The BAB strategy of Frazzini and Pedersen (2014) exploits the price distortions caused by leverage-constrained investors on low- and high-beta stocks. The variance factor tracks the realized variance of the S&P 500. The global carry and TS momentum factors of Kojien et al. (2018) and Moskowitz et al. (2012) invest in assets with high carry and positive 12-month returns across international asset classes. TS momentum departs from traditional cross-sectional momentum, which only invests in assets with past returns higher than the cross-sectional average.

¹⁸ We define the variance and straddle factors as short positions to obtain positive premia (i.e., these short positions perform poorly in bad times when realized variance is high (Bakshi and Kapadia, 2003)).

Table 2
Summary Statistics for the Hedge Fund Factors.

This table provides summary statistics for the factors used in the construction of the nine hedge fund models. Panel A reports the statistics for the US equity market and the standard factors used for performance evaluation. The size, value, momentum, investment, and profitability factors are computed using US equity data. The global value and momentum factors are constructed across multiple international asset classes. The term and default factors are computed using US bond data. The bond, commodity, and currency straddles are computed using option data on bonds, commodities, and currencies. Panel B reports the statistics for the additional hedge fund factors. The illiquidity and betting-against-beta (BAB) factors are computed using US equity data. The variance factor is computed using index option data on the S&P500. The carry and time-series (TS) momentum factors are constructed across multiple international asset classes. We define the straddle and variance factors as short positions to obtain positive premia. We report the annualized mean and standard deviation, skewness, kurtosis, and quantiles at 10% and 90% of the excess returns of the factors. The statistics are computed using monthly data between January 1994 and December 2020.

	Panel A: Market and Standard Factors					
	Moments				Quantiles	
	Mean	Std Dev.	Skewness	Kurtosis	10%	90%
Market	8.81	15.48	-0.64	4.26	-5.15	6.02
Size	1.61	10.83	0.38	7.54	-3.49	3.69
Value	0.20	10.88	0.05	5.84	-3.30	3.47
Momentum	4.67	17.15	-1.42	12.88	-5.07	5.50
Investment	2.11	7.04	0.67	5.09	-2.14	2.70
Profitability	3.77	9.38	-0.47	13.49	-2.03	2.90
Global Value	1.30	6.19	-0.64	12.25	-1.72	1.76
Global Momentum	3.26	7.55	-0.30	5.46	-2.25	2.69
Term	-0.18	0.89	-0.03	4.22	-0.35	0.29
Default	0.01	0.77	1.90	17.66	-0.20	0.19
Straddle on Bonds	16.00	57.86	-1.85	8.99	-19.68	17.73
Straddle on Commodities	2.99	50.38	-1.32	6.07	-19.71	15.74
Straddle on Currencies	11.27	69.02	-1.56	6.64	-23.36	20.15

	Panel B: Additional Factors					
	Moments				Quantiles	
	Mean	Std Dev.	Skewness	Kurtosis	10%	90%
Illiquidity	6.52	12.97	-0.27	4.49	-3.60	4.99
BAB	8.58	13.77	-0.36	6.10	-3.58	5.21
Variance	38.10	23.80	-4.57	29.76	-1.03	7.61
Carry	6.85	4.96	0.01	3.98	-1.10	2.34
TS Momentum	11.11	12.56	0.16	3.16	-3.54	5.71

above 0.5 (in absolute value). Table 3 provides a complete list of the nine models with the factors they include (see the appendix for details on the data sources of the factors).

4.3. Misspecification statistics

Before moving to the main empirical results, we examine the misspecification of the nine models. An intuitive statistic is the adjusted R^2 of the fund-by-fund regression in Equation (4). A high R^2 is a signal of low misspecification because the omitted factors must have implausibly high premia to substantially affect the fund return decomposition. Table 4 shows that the average R^2 is relatively low across all models—the values range between 20.4% and 31.0%, leaving plenty of room for omitted factors.

In theory, a correct model could deliver a low R^2 if hedge funds take large idiosyncratic positions to exploit their private information. In this case, a low R^2 might capture skill, not misspecification (Titman and Tiu, 2011). To address this issue, we compute the criterion of Gagliardini et al. (2019). As n and T converge to infinity, it is positive with probability one if (i) the model is misspecified, and (ii) at least one omitted factor is strong (see the appendix). We find that this criterion is systematically positive. This result confirms that all the models are misspecified. It also validates our estimation assumption that one of the omitted factors is strong.

While misspecified, the nine models are likely to produce different return decompositions. Table 4 provides preliminary evidence consistent with this view. For each model k , we report the relative importance

of each component of the average fund return: (i) the alpha component $\hat{a}_i^k$, (ii) the beta component due to the market computed as $\hat{b}_{i,m}^k = \hat{b}_{i,m}^k \hat{\lambda}_m$, and (iii) the beta component due to the non-market (nm) factors (all factors but the market) computed as $\hat{b}_{i,I_{nm}}^k = \hat{b}_{i,I_{nm}}^k \hat{\lambda}_{I_{nm}}^k$. For instance, the relative importance of the alpha component is equal to 47.1% on average under the Carhart model but drops to 18.0% under the JKKT model. Interpreting these differences requires formal comparison tests—a point we examine below.

Looking at the population of US equity mutual funds, we see that the average R^2 is substantially higher (above 80%). In addition, all models lead to the same conclusion that mutual fund returns are driven by a single component—the equity market. In short, these results confirm that accounting for misspecification is crucial for hedge funds but not for mutual funds.

5. Empirical results

5.1. Hedge fund return decomposition

5.1.1. Formal model comparison

We begin the empirical analysis with the formal model comparison. This analysis determines whether some models do a better job than the CAPM at capturing the alternative strategies followed by hedge funds. Applying our methodology, we compare the alpha distributions under each proposed model and the CAPM. We explicitly test for differences in the mean and standard deviation, proportions of negative-

Table 3**The Set of Hedge Fund Models.**

This table summarizes the set of nine hedge fund models chosen for the empirical analysis. This set includes the CAPM as reference model as well as three distinct groups. The first group includes the standard models used in previous work, namely the Carhart, Five-Factor, Fung-Hsieh, and Asness-Moskowitz-Pedersen (AMP) models. The second group includes the two machine learning models of Kozak, Nagel, and Shantosh (KNS) formed with five characteristic-based equity portfolios and five principal components of these portfolios. The final group includes two models formed with the additional hedge fund factors. The first one is the model of Joenväärä et al. (2021) (JKKT). The second one combines the factors proposed by Carhart et al. (2014) and Pedersen (2018) (CP).

Model	List of Included Factors
Reference Model	
CAPM	Market
Standard Models	
Carhart	Market, Size, Value, Momentum
Five-Factor	Market, Size, Value, Investment, Profitability
Fung-Hsieh	Market, Size, Term, Default, Straddles (Bonds, Commodities, Currencies)
AMP	Market, Global Value and Momentum
Machine Learning Models	
KNS1	Market, Five Characteristic-Based Equity Factors
KNS2	Market, Five Principal Component Equity Factors
Additional Models	
JKKT	Market, Size, Value, Momentum, Illiquidity, BAB, TS Momentum
CP	Market, Size, Illiquidity, BAB, Variance, Carry, TS Momentum

Table 4**Model Misspecification Statistics.**

This table provides misspecification statistics for the CAPM, the four standard models (Carhart, Five-Factor, Fung-Hsieh, AMP), the two machine learning models (KNS1, KNS2), and the two models with the additional factors (JKKT and CP). For each model, we measure the relative importance of the three components of the average fund return: (i) the alpha component (Alpha), (ii) the beta component due to the market (Beta Mkt), and the beta component due to the non-market factors (Beta Non-Mkt). These proportions, which are averaged across all funds, sum up to 100%. We also compute the average adjusted R^2 of the time-series regression of the fund return on the factor returns. We conduct this analysis for the entire hedge fund population (first four columns) and for the entire mutual fund population (last four columns).

	Hedge Funds				Mutual Funds			
	Relative Importance (%)				Relative Importance (%)			
	Alpha	Beta Mkt	Beta Non-Mkt	R^2	Alpha	Beta Mkt	Beta Non-Mkt	R^2
CAPM	52.75	47.25	0.00	20.43	-15.23	115.23	0.00	81.34
Carhart	47.14	46.60	6.26	25.28	-17.16	110.57	6.58	88.98
Five-Factor	49.04	46.65	4.31	24.85	-16.38	110.27	6.11	88.99
Fung-Hsieh	54.10	40.56	5.34	30.24	-14.96	107.78	7.18	85.88
AMP	47.08	47.37	5.55	24.36	-15.37	115.01	0.36	84.55
KNS1	52.52	48.24	-0.76	24.47	-17.41	114.58	2.84	83.40
KNS2	64.01	51.03	-15.04	26.39	-6.27	115.99	-9.73	87.09
JKKT	17.97	44.28	37.75	31.01	-20.01	110.20	9.81	89.65
CP	6.64	41.19	52.17	30.26	-24.33	109.06	15.27	87.15

and positive-alpha funds, and quantiles at 10% and 90%. To compute the standard deviation of the estimated differences, we replace T with $T_\chi = \frac{1}{n_\chi} \sum_{i=1}^n \mathbf{1}_i^\chi T_i$, where T and T_χ are equal to 324 and 125 observations. As a result, we account for the increased noise due to the unbalanced nature of the hedge fund panel.

The comparison tests reported in Table 5 reveal several new insights. We observe a striking similarity between the CAPM and the four standard models (Carhart, Five-Factor, Fung-Hsieh, and Asness-Moskowitz-Pedersen). Only one of the 24 characteristic differences is statistically significant. Properly accounting for misspecification is key when interpreting these differences. Suppose that we naively use the convergence rate of $1/\sqrt{n_\chi}$, instead of the appropriate rate of $1/\sqrt{T_\chi}$ under misspecification. In this case, we find that 75% of the 24 characteristic differences are significant at the 5% level. Applying the correct

procedure, we conclude that the standard models are no better than the CAPM at capturing alternative hedge fund strategies.

Similarly, the machine learning models produce the same alpha distributions as the CAPM. Whereas these models do a great job at explaining the average returns of characteristic-based equity portfolios (Kozak et al., 2020), they are not trained on strategies beyond the equity space. Their limited success in the hedge fund population highlights the importance of accounting for other asset classes, such as bonds, currencies, and commodities.

The only exceptions are the JKKT and CP models. The tests reveal highly significant differences relative to the CAPM. These differences hold after replacing the CAPM with any of the standard and machine learning models. They also hold when accounting for factor trading costs using estimates from previous studies (see the appendix). In short,

Table 5**Formal Model Comparisons.**

This table compares the CAPM with the four standard models (Carhart, Five-Factor, Fung-Hsieh, AMP), the two machine learning models (KNS1, KNS2), and the two models with the additional factors (JKKT and CP). We compute the differences in characteristics between the cross-sectional distributions of the alpha components under the CAPM and each model. We report the differences in the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated differences. ***, **, * indicate that the null hypothesis of equal characteristics is rejected at the 1%, 5%, and 10% levels.

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Carhart	-0.31 (0.40)	-0.19 (0.31)	2.27 (2.50)	-2.27 (2.50)	-0.02 (0.40)	-0.60* (0.34)
Five-Factor	-0.21 (0.46)	0.02 (0.34)	2.41 (3.01)	-2.41 (3.01)	-0.01 (0.50)	-0.18 (0.40)
Fung-Hsieh	0.08 (0.60)	-0.20 (0.35)	0.52 (3.88)	-0.52 (3.88)	0.28 (0.56)	-0.15 (0.47)
AMP	-0.31 (0.42)	0.06 (0.37)	2.50 (2.44)	-2.50 (2.44)	-0.17 (0.37)	-0.34 (0.45)
KNS1	-0.01 (0.39)	0.43 (0.29)	1.59 (1.78)	-1.59 (1.78)	-0.26 (0.29)	0.19 (0.38)
KNS2	0.63 (0.49)	0.20 (0.31)	-2.47 (2.65)	2.47 (2.65)	0.58 (0.41)	0.65 (0.47)
JKKT	-1.93*** (0.67)	0.41 (0.50)	14.76*** (4.01)	-14.76*** (4.01)	-2.32*** (0.66)	-1.98*** (0.68)
CP	-2.56*** (0.76)	2.13*** (0.50)	19.82*** (4.27)	-19.82*** (4.27)	-4.22*** (0.73)	-1.42* (0.76)

Table 6**Decomposition of Average Fund Returns.**

This table shows the decomposition of average fund returns under the CAPM, the four standard models (Carhart, Five-Factor, Fung-Hsieh, AMP), the two machine learning models (KNS1, KNS2), and the two models with the additional factors (JKKT and CP). Panel A reports the characteristics of the cross-sectional distribution of the alpha components under each model. We report the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B reports the characteristics of the cross-sectional distribution of beta components under each model.

Panel A: Distribution of the Alpha Components						
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
CAPM	2.93 (0.94)	7.01 (0.48)	27.15 (5.49)	72.85 (5.49)	-3.95 (0.67)	10.06 (0.67)
Carhart	2.62 (0.84)	6.82 (0.31)	29.42 (5.08)	70.58 (5.08)	-3.97 (0.57)	9.46 (0.55)
Five-Factor	2.72 (0.89)	7.03 (0.31)	29.55 (5.23)	70.45 (5.23)	-3.96 (0.63)	9.87 (0.54)
Fung-Hsieh	3.01 (0.74)	6.81 (0.29)	27.66 (3.60)	72.34 (3.60)	-3.67 (0.47)	9.90 (0.51)
AMP	2.62 (0.92)	7.08 (0.28)	29.65 (5.62)	70.35 (5.62)	-4.12 (0.59)	9.71 (0.54)
KNS1	2.92 (0.87)	7.44 (0.42)	28.73 (5.13)	71.27 (5.13)	-4.21 (0.63)	10.24 (0.60)
KNS2	3.56 (0.86)	7.21 (0.37)	24.68 (4.33)	75.32 (4.33)	-3.37 (0.55)	10.70 (0.62)
JKKT	1.00 (0.73)	7.42 (0.31)	41.90 (4.80)	58.10 (4.80)	-6.27 (0.60)	8.07 (0.38)
CP	0.37 (0.86)	9.15 (0.40)	46.97 (5.31)	53.03 (5.31)	-8.17 (0.77)	8.64 (0.44)

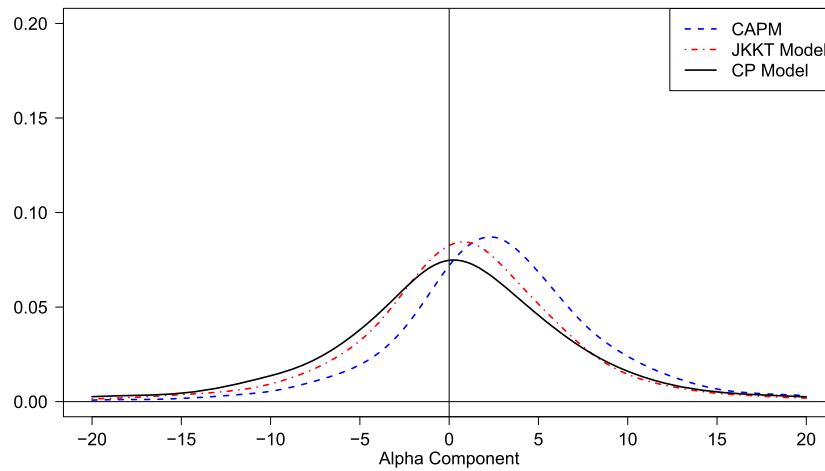
Panel B: Distribution of the Beta Components						
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
CAPM	2.62 (0.83)	4.37 (0.67)	22.54 (9.57)	77.46 (9.57)	-0.71 (0.61)	8.01 (0.73)
Carhart	2.94 (0.81)	4.30 (0.52)	17.99 (5.68)	82.01 (5.68)	-0.65 (0.42)	8.20 (0.72)
Five-Factor	2.83 (0.84)	4.48 (0.51)	20.61 (5.62)	79.39 (5.62)	-1.06 (0.41)	8.32 (0.75)
Fung-Hsieh	2.55 (0.78)	4.39 (0.52)	22.58 (5.98)	77.42 (5.98)	-1.23 (0.44)	7.96 (0.67)
AMP	2.94 (0.84)	4.66 (0.49)	18.98 (5.41)	81.02 (5.41)	-0.96 (0.43)	8.56 (0.73)
KNS1	2.64 (0.80)	5.14 (0.57)	23.04 (5.71)	76.96 (5.71)	-1.37 (0.56)	8.38 (0.86)
KNS2	2.00 (0.82)	4.47 (0.49)	27.53 (7.96)	72.47 (7.96)	-1.61 (0.59)	7.28 (0.69)
JKKT	4.56 (0.83)	5.90 (0.42)	13.90 (2.20)	86.10 (2.20)	-0.50 (0.28)	11.12 (0.74)
CP	5.19 (0.93)	7.64 (0.49)	15.91 (2.17)	84.09 (2.17)	-1.17 (0.35)	12.78 (1.11)

the JKKT and CP models are better equipped to capture alternative strategies and sharpen the fund return decomposition.

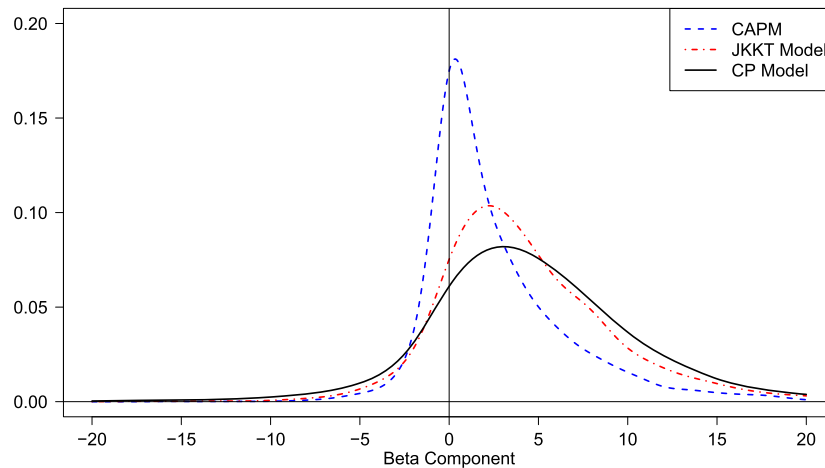
5.1.2. Magnitude of the alpha and beta components

We now turn to the return decomposition for the entire population. For each model k , we apply our methodology to infer the main characteristics of (i) the distribution of the alpha components $\hat{\phi}_{ac}^k$, and (ii) the distribution of the beta components $\hat{\phi}_{bc}^k$. We report these results in Table 6.

Both the CAPM and the standard models produce large alpha components. Panel A shows that the average alpha clusters around 2.8% per year, and the proportion of positive-alpha funds is always above 70%. These results are in line with the previous literature which finds that hedge funds deliver superior performance. Another takeaway from these models is that hedge funds are only exposed to market risk. Panel B shows that the average beta component equals 2.6% under the CAPM and remains unchanged as we include profitability, bond, or straddle factors. In short, hedge funds deliver high alphas to investors while being immune to alternative sources of risk.



(a) Distributions of the Alpha Components



(b) Distributions of the Beta Components

Fig. 2. Distributions of the Alpha and Beta Components.

This figure compares the average return decomposition obtained with the two models with the additional factors (JKKT and CP) and the CAPM. Panel A plots the cross-sectional distributions of the alpha components (annualized) under the three models. Panel B repeats the analysis for the beta components.

The JKKT and CP models reverse these conclusions. For instance, the CP model delivers an average beta component of 5.2%, which is twice as large as under the CAPM. At the same time, it brings the average alpha down to 0.4% per year and the proportion of positive-alpha funds down to 53.0%. These results highlight the importance of the additional factors included in the JKKT and CP models. As discussed by Asness et al. (2013) and Pedersen (2015), there is ample anecdotal evidence that hedge funds hold illiquid assets, take levered positions, trade equity options, buy cheap assets with high carry, and follow trends in asset prices.

To visualize these results, we plot in Fig. 2 the densities of the two components for the CAPM, the JKKT model, and the CP model. In line with our illustrative example in Fig. 1, these two models capture the premia of alternative strategies and thus produce (i) a shift of the alpha distribution towards zero and (ii) a shift of the beta distribution away from zero.

5.1.3. Economic importance of each factor

Our previous analysis highlights the key role played by the additional factors. To measure their economic importance separately, we

break down the total beta component obtained with the CP model into the respective contributions due to market, size, illiquidity, BAB, variance, carry, and TS momentum. Using these estimated quantities, we then apply our methodology to infer the main characteristics of the beta distribution $\phi_{bc,j}^k$ for each factor.

Table 7 shows that the equity market is the most prevalent source of risk as 78% of the funds have positive market betas. On average, the market contribution equals 2.3% per year, representing 44% of the total beta component. We also find that a majority of funds load positively on each of the five additional strategies. They are therefore widely used by hedge funds to boost their returns. Interestingly, individual funds do not load on all of them simultaneously. As shown in Panel B, the pairwise correlations between the factor components only range between -18% and 14%.

TS momentum, variance, and carry are the most important additional factors. Their average contributions are all statistically significant and equal to 1.1%, 0.8%, and 0.4% per year. In the top decile of funds with the highest exposures, these contributions reach 4.5%, 4.5%, and 2.7% per year. The unique feature of the CP model is to include all three factors, which explains its edge over the JKKT model. It consistently

Table 7**Economic Importance of the Hedge Fund Factors.**

This table measures the economic importance of each factor in the CP model as a driver of average fund returns. Panel A reports the characteristics of the cross-sectional distribution of the beta components due to each factor. We report the annualized mean and standard deviation, the proportions of funds with negative and positive contributions, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B reports the cross-sectional correlation between the beta components for each pair of factors.

	Panel A: Distribution of the Beta Components for Each Factor					
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Market	2.29 (1.27)	3.84 (1.19)	21.62 (6.13)	78.38 (6.13)	-0.55 (0.15)	6.99 (2.05)
Size	0.23 (0.26)	1.01 (0.32)	38.23 (4.26)	61.77 (4.26)	-0.31 (0.03)	1.02 (0.53)
Illiquidity	0.05 (0.07)	1.26 (0.29)	44.94 (3.25)	55.06 (3.25)	-0.75 (0.24)	0.85 (0.38)
Betting Against Beta	0.35 (0.25)	2.22 (0.69)	35.14 (3.80)	64.86 (3.80)	-1.30 (0.44)	2.23 (0.90)
Variance	0.78 (0.25)	5.17 (0.70)	36.74 (1.89)	63.26 (1.89)	-3.08 (0.54)	4.53 (0.87)
Carry	0.43 (0.17)	2.56 (0.46)	39.21 (2.38)	60.79 (2.38)	-1.89 (0.37)	2.73 (0.51)
Time-Series Momentum	1.06 (0.32)	4.03 (0.78)	42.86 (1.51)	57.14 (1.51)	-1.35 (0.34)	4.49 (0.72)

	Panel B: Correlations Between the Beta Components					
	Size	Illiquidity	BAB	Variance	Carry	TS Mom
Market	-0.08	-0.01	-0.01	-0.03	-0.02	-0.18
Size		-0.07	-0.00	-0.06	0.06	0.05
Illiquidity			-0.09	0.14	-0.02	0.08
Betting Against Beta				-0.07	0.03	-0.05
Variance					0.04	-0.09
Carry						-0.13

delivers lower average alphas and lower proportions of positive-alpha funds (both in the population and for all investment categories). For this reason, we primarily focus on the CP model for the rest of our analysis.

5.2. Hedge fund investment styles

5.2.1. A closer look at equity, macro, and arbitrage funds

Whereas the CP model includes factors that plausibly capture several hedge fund strategies, it is not designed for a particular style. Therefore, an important question is whether the superiority of the CP model holds across categories. To examine this issue, we re-estimate the alpha and beta distributions for equity, macro, and arbitrage funds under the CAPM, the JKKT model, and the CP model. The other models deliver the same results as the CAPM and are thus not shown.

Table 8 confirms the strong differences between the CP model and the CAPM. Among equity and arbitrage funds, the average alphas under the CP model are equal to 0.6% and 0.9% per year (versus 2.4% and 2.8% for the CAPM). In the macro category, the difference is even larger as the majority of funds deliver a negative alpha equal to -0.4% on average (versus 3.7% for the CAPM). For completeness, we examine multi-strategy funds and funds of funds which could be more difficult to model given their diversity. In both categories, the appendix shows that the CP model still delivers a sharp reduction in alphas.

These results suggest that the additional factors in the CP model transcend style boundaries. Carry provides a good illustration of this phenomenon. This strategy invests in cheap assets with a high difference between spot and forward prices. As discussed by Pedersen (2015), carry is routinely implemented by hedge funds, regardless of their investment styles. It is the case when they buy value stocks, high interest rate currencies, or backward commodities. Consistent with this analysis, Table 9 reveals that carry matters for all three categories. Its average beta component is always positive and ranges between 0.34% (equity) and 0.56% (arbitrage) per year.

Table 9 also shows that variance transcends style boundaries. Its average return contribution among equity and arbitrage funds is equal to 1.1% per year. The economic importance of variance reflects the option strategies followed by these funds. For instance, mortgage, fixed income volatility, and merger arbitrage activities all involve taking short option positions (Duarte et al., 2006; Mitchell and Pulvino, 2001). In addition, the variance factor captures unexpected increases in asset correlations (Driessen et al., 2009). As such, it signals crisis times when equity and

arbitrage funds suffer from less effective hedging strategies and tighter funding constraints (Buraschi et al., 2014b).

At the same time, some factors in the CP model cater to specific styles. For instance, TS momentum plays a key role for macro funds—its average contribution reaches 3.10% per year. This result is consistent with the intuition that global funds rely on past returns to determine their asset allocation and exploit trends caused by behavioral biases, frictions, or slow-moving capital.¹⁹ Another example is BAB which primarily matters for arbitrage funds. These funds extensively use leverage to exploit price distortions in capital markets (Ang et al., 2011). It is therefore plausible that some of these distortions originate from the leverage constraints faced by traditional investors and captured by the BAB strategy.

5.2.2. The heterogeneity across funds

The return decomposition within styles in Table 8 uncovers a large fund heterogeneity. Starting with the alpha distribution, we see that the CP model produces a cross-sectional volatility between 6.8% (arbitrage) and 11.0% (macro) per year. Similarly, the beta distribution covers a wide interval as its volatility ranges between 4.9% (arbitrage) and 9.6% (macro) per year. These values are similar to those obtained in the entire population—Table 6 reports a volatility of 9.2% and 7.6% per year for the alpha and beta distributions.²⁰ The rationale for forming groups is to summarize the information conveyed by factor models and identify funds with similar performance and risk levels. Meeting this objective is however difficult for hedge funds given their diverse strategies.

It might be surprising that the CP model always produces a higher dispersion in alphas than the CAPM. A priori, the CAPM should generate a higher variance because its alpha absorbs the dispersion due to the omitted factors (the term $2\sigma_b^* \lambda^2$ in our example in Fig. 1). Whereas

¹⁹ Managed futures funds (a subset of macro funds) are well known for following trends. Unsurprisingly, the appendix shows that the average contribution of TS momentum for these funds is even higher (3.7% per year).

²⁰ This heterogeneity may partly be driven by misclassification. First, data providers use different style classifications that need to be reconciled when creating the aggregate database (see the appendix). Second, the methodology used by each provider (e.g., clustering) and the information provided by funds could introduce classification noise. Third, the reported styles are not frequently updated and may therefore not always reflect the actual fund strategies.

Table 8

Decomposition of Average Fund Returns – Investment Categories.

This table shows the decomposition of average fund returns under the CAPM and the two models with the additional factors (JKKT and CP) across investment styles. Panel A reports the characteristics of the cross-sectional distributions of the alpha and beta components across equity funds (long-short, market neutral). We report the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panels B and C repeat the analysis for macro funds (global macro, managed futures) and arbitrage funds (relative value, event driven).

	Moments (Ann.)		Panel A: Equity Funds Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Distribution of the Alpha Components						
CAPM	2.40 (1.06)	7.09 (0.48)	30.23 (6.31)	69.77 (6.31)	-4.52 (0.83)	9.54 (0.96)
JKKT	1.12 (0.81)	6.98 (0.31)	42.98 (5.48)	57.02 (5.48)	-5.73 (0.75)	7.77 (0.66)
CP	0.58 (0.96)	9.08 (0.49)	47.38 (5.46)	52.62 (5.46)	-7.84 (1.02)	8.53 (0.79)
Distribution of the Beta Components						
CAPM	4.18 (1.16)	5.19 (0.70)	12.94 (8.25)	87.06 (8.25)	-0.14 (0.74)	10.66 (1.39)
JKKT	5.46 (1.18)	5.75 (0.54)	10.59 (2.37)	89.41 (2.37)	-0.10 (0.43)	12.19 (1.41)
CP	6.00 (1.31)	7.58 (0.61)	13.69 (2.77)	86.31 (2.77)	-0.75 (0.56)	13.71 (1.77)
	Moments (Ann.)		Panel B: Macro Funds Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Distribution of the Alpha Components						
CAPM	3.71 (1.72)	8.04 (0.78)	25.73 (6.47)	74.27 (6.47)	-4.26 (0.89)	12.41 (2.14)
JKKT	-0.14 (1.42)	9.14 (0.65)	51.52 (7.66)	48.48 (7.66)	-9.25 (1.43)	8.70 (0.75)
CP	-0.39 (1.65)	11.04 (0.70)	51.83 (7.02)	48.17 (7.02)	-10.91 (1.58)	10.26 (1.00)
Distribution of the Beta Components						
CAPM	1.01 (1.08)	3.70 (0.68)	43.44 (21.78)	56.56 (21.78)	-1.76 (1.29)	5.44 (0.66)
JKKT	4.87 (1.11)	7.49 (0.74)	20.32 (3.33)	79.68 (3.33)	-1.57 (0.44)	12.86 (1.30)
CP	5.11 (1.33)	9.62 (0.79)	23.74 (3.89)	76.26 (3.89)	-2.60 (0.63)	14.18 (1.35)
	Moments (Ann.)		Panel C: Arbitrage Funds Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Distribution of the Alpha Components						
CAPM	2.81 (1.24)	5.61 (0.44)	24.74 (9.71)	75.26 (9.71)	-2.79 (1.31)	8.67 (0.63)
JKKT	1.97 (0.89)	5.70 (0.29)	31.03 (8.00)	68.97 (8.00)	-3.66 (0.92)	7.60 (0.50)
CP	0.87 (1.06)	6.81 (0.44)	41.64 (9.30)	58.36 (9.30)	-5.48 (1.20)	7.62 (0.49)
Distribution of the Beta Components						
CAPM	2.31 (0.98)	3.02 (0.66)	13.63 (8.51)	86.37 (8.51)	-0.08 (0.45)	5.88 (1.22)
JKKT	3.14 (0.81)	3.57 (0.42)	11.60 (3.04)	88.40 (3.04)	-0.09 (0.32)	7.28 (1.03)
CP	4.25 (0.98)	4.90 (0.56)	10.86 (2.29)	89.14 (2.29)	-0.10 (0.33)	9.42 (1.37)

this effect is at play, it is more than offset by the reduction in variance due to the negative cross-sectional correlation between the alpha and beta components. This correlation arises because poorly performing funds load aggressively on alternative strategies—possibly to hide their lack of skill. To elaborate, consider the worst-performing macro funds under the CP model. These funds load heavily on the five additional factors to boost their returns. Unreported results show that their average beta component is three times larger than in the population (16.1% versus 5.2% per year). By controlling for these factor exposures, the CP model uncovers the strong underperformance of these funds—the 10%-quantile equals -8.2% per year (versus -4.0% only under the CAPM).

5.2.3. Benchmarking using style-specific models

A natural question is whether models formed with style-specific factors can sharpen the decomposition of hedge fund returns. Our previous analysis reveals the existence of a trade-off. On the one hand, these models may capture the prevalent strategies in each style and be less prone to misspecification. On the other hand, the observed heterogeneity calls against using rigid style models which assume that all funds follow similar strategies. Heterogeneity may also increase the list of candidate style factors to choose from and thus the risk of data mining.

In this context, the common approach of peer benchmarking is problematic for hedge funds. This approach uses as benchmark a style index

that includes all funds in a given investment category (e.g., Hunter et al., 2014; Buraschi et al., 2014a). Using a style index produces substantial misspecification because it imposes a constant beta component for all funds with the same style—a restriction that is inconsistent with the heterogeneity reported in Table 8. It implies that any fund with a beta component lower than average is mechanically credited with a lower alpha. Another issue is that hedge fund style indices are not investible given the numerous constraints that prevent investors from forming diversified hedge fund portfolios.²¹

The above trade-off suggests that models with style-specific factors are likely to perform better on homogeneous categories with well-defined strategies. For instance, Duarte et al. (2006) show the benefits of using these models for explaining the returns of fixed income arbitrage funds (e.g., swap spread, yield curve arbitrage). In contrast, finding appropriate style factors to cover the three broad categories examined here is more challenging. As a simple illustration, we consider a style-based version of the CP model in which we replace the

²¹ Investability is less problematic for mutual funds. As noted by Hunter et al. (2014): “Even the least sophisticated investor always has a fallback strategy of equally-weighting (or value-weighting) all funds in the group every period; this tradeable strategy is quite simple.”

Table 9**Economic Importance of the Hedge Fund Factors – Investment Categories.**

This table measures the economic importance of each factor in the CP model as a driver of average fund returns across investment styles. Panel A reports the characteristics of the cross-sectional distributions of the beta components due to each factor across equity funds (long-short, market neutral). We report the annualized mean and standard deviation, the proportions of funds with negative and positive contributions, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B and C repeat the analysis for macro funds (global macro, managed futures) and arbitrage funds (relative value, event driven).

	Panel A: Equity Funds					
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Market	3.67 (2.07)	4.53 (1.22)	14.54 (8.16)	85.46 (8.16)	-0.19 (0.61)	9.33 (3.18)
Size	0.42 (0.57)	1.32 (0.42)	32.68 (11.70)	67.32 (11.70)	-0.35 (0.11)	1.65 (1.04)
Illiquidity	0.14 (0.10)	1.49 (0.35)	44.13 (2.66)	55.87 (2.66)	-0.85 (0.25)	1.27 (0.48)
Betting Against Beta	0.25 (0.21)	2.66 (0.80)	38.48 (3.90)	61.52 (3.90)	-1.71 (0.55)	2.55 (1.05)
Variance	0.67 (0.25)	5.06 (0.70)	38.58 (2.19)	61.42 (2.19)	-3.12 (0.54)	4.46 (0.85)
Carry	0.34 (0.17)	2.80 (0.46)	44.78 (2.67)	55.22 (2.67)	-2.11 (0.41)	2.73 (0.52)
Time-Series Momentum	0.51 (0.17)	2.83 (0.55)	42.08 (2.04)	57.92 (2.04)	-1.56 (0.46)	3.05 (0.74)

	Panel B: Macro Funds					
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Market	1.18 (0.62)	3.47 (1.10)	35.11 (4.40)	64.89 (4.40)	-1.40 (0.35)	4.87 (1.62)
Size	0.07 (0.09)	0.83 (0.20)	50.59 (9.02)	49.41 (9.02)	-0.43 (0.28)	0.64 (0.20)
Illiquidity	-0.08 (0.10)	1.26 (0.31)	54.07 (3.37)	45.93 (3.37)	-0.90 (0.37)	0.75 (0.28)
Betting Against Beta	0.21 (0.19)	1.99 (0.65)	40.83 (4.63)	59.17 (4.63)	-1.30 (0.47)	1.84 (0.76)
Variance	0.22 (0.35)	6.56 (0.87)	49.41 (3.11)	50.59 (3.11)	-4.73 (0.92)	4.55 (0.76)
Carry	0.41 (0.26)	2.79 (0.57)	41.89 (3.33)	58.11 (3.33)	-2.43 (0.58)	3.23 (0.64)
Time-Series Momentum	3.10 (1.11)	5.83 (1.23)	24.43 (3.26)	75.57 (3.26)	-0.73 (0.12)	10.22 (2.20)

	Panel C: Arbitrage Funds					
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Market	1.68 (0.90)	2.52 (0.75)	16.96 (6.38)	83.04 (6.38)	-0.19 (0.21)	4.40 (1.44)
Size	0.16 (0.20)	0.60 (0.17)	32.82 (8.47)	67.18 (8.47)	-0.11 (0.04)	0.56 (0.45)
Illiquidity	0.07 (0.12)	0.87 (0.18)	36.89 (7.20)	63.11 (7.20)	-0.40 (0.06)	0.60 (0.32)
Betting Against Beta	0.61 (0.39)	1.76 (0.58)	25.35 (4.61)	74.65 (4.61)	-0.68 (0.14)	2.15 (0.95)
Variance	1.47 (0.42)	3.34 (0.55)	21.90 (2.39)	78.10 (2.39)	-0.78 (0.08)	4.59 (0.97)
Carry	0.56 (0.20)	1.91 (0.34)	29.67 (3.66)	70.33 (3.66)	-0.86 (0.17)	2.37 (0.52)
Time-Series Momentum	-0.30 (0.19)	1.53 (0.37)	62.12 (4.12)	37.88 (4.12)	-1.64 (0.57)	0.82 (0.17)

carry and TS momentum factors with their style-specific counterparts. For each category, the appendix shows that these replacements decrease the ability of the CP model to capture hedge fund returns.

5.3. The implications of fund heterogeneity

5.3.1. Hedge fund selection

The large heterogeneity in alphas implies that investors face substantial uncertainty in their fund selection process. One possible approach for reducing this uncertainty is to use fund characteristics to improve the detection of positive-alpha funds. To examine this issue, we focus on a sophisticated investor who evaluates performance using the CP model and has access to two sets of characteristics. The first set proxies for managerial incentives and includes management fees, performance fees, and a high-water mark dummy. The second set proxies for managerial flexibility and includes a lockup period dummy and a notice period dummy. For each characteristic, we sort funds in two groups and apply our methodology to estimate their respective alpha distributions.

Panel A of Table 10 reports the results for the first set of characteristics. We find that sorting funds on each proxy for managerial incentives increases the likelihood of selecting positive-alpha funds. For instance, close to 60.0% of the funds that charge high performance fees produce a positive alpha equal to 1.1% per year on average. In contrast, the average alpha is negative among funds with low performance fees (-1.5%) as only 39.2% of them deliver positive alphas.

Panel B documents similar patterns for lockup and notice periods. Funds with higher flexibility tend to perform better, possibly because they can invest in illiquid assets and exploit arbitrage opportunities that

take time to be profitable. In short, all five characteristics are useful for partially discriminating between inferior and superior funds. This result emphasizes the importance of managerial incentives and flexibility as determinants of fund performance, which is in line with previous studies by Agarwal et al. (2009), Aragon (2007), and Joenväärä et al. (2021).

5.3.2. Models of active management

The observed heterogeneity has also implications for the models of Berk and Green (2004) and Gârleanu and Pedersen (2018)—two popular equilibrium models of active management. In the model of Berk and Green (2004), skilled funds have bargaining power because they are in short supply. As investors compete for performance, they drive the alphas of all funds towards zero and eliminate any heterogeneity (after an adjustment period due to learning). This prediction holds quite well for mutual funds—Barras et al. (2022) find that the standard deviation of the alpha distribution only equals 1.4% per year. However, it is at odds with the alpha volatility of 9.2% per year obtained with the CP model.

In contrast, the model of Gârleanu and Pedersen (2018) predicts that alphas vary across funds in equilibrium. In this model, skilled funds deliver positive alphas because they must compensate investors for the costs of searching funds. At the same time, unskilled funds deliver negative alphas as they charge fees to unsophisticated investors. Modeling the equilibrium of the hedge fund industry using search models is intuitive given the complex process for evaluating hedge funds (Lhabitant, 2007). If the results in Table 6 reflect equilibrium outcomes, an open

Table 10**Fund Characteristics and Alphas.**

This table examines how the alpha component varies across groups of funds sorted on fund characteristics. Panel A reports the characteristics of the cross-sectional distribution of the alpha component for groups sorted on proxies of managerial incentives, which are management fees, performance fees, and a dummy that takes a value of one if the fund has a high water mark provision. We set the cutoff levels at the usual 2% for management fees and 20% for performance fees. We report the annualized mean and standard deviation, the proportions of funds with negative and positive contributions, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B repeats the analysis using proxies of managerial flexibility, which are two dummies that take a value of one if the fund imposes a lockup period or a notice period.

	Panel A: Managerial Incentives					
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low Management Fees	0.01 (0.81)	8.03 (0.38)	49.04 (5.91)	50.96 (5.91)	-7.63 (0.84)	7.39 (0.48)
High Management Fees	1.28 (1.27)	11.41 (0.59)	41.72 (4.99)	58.28 (4.99)	-10.08 (1.32)	11.31 (0.67)
Low Performance Fees	-1.53 (0.98)	7.30 (0.41)	60.77 (7.43)	39.23 (7.43)	-8.17 (1.18)	5.44 (0.50)
High Performance Fees	1.11 (0.90)	9.70 (0.41)	41.44 (5.08)	58.56 (5.08)	-8.20 (0.80)	9.85 (0.51)
No High Water Mark	-1.44 (1.35)	9.97 (0.74)	58.38 (5.48)	41.62 (5.48)	-10.23 (1.20)	7.48 (0.83)
High Water Mark	1.17 (0.87)	8.82 (0.43)	41.55 (5.58)	58.45 (5.58)	-7.39 (0.89)	9.36 (0.46)

	Panel B: Managerial Flexibility					
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
No Lockup Period	0.03 (0.85)	9.23 (0.39)	48.77 (5.27)	51.23 (5.27)	-8.52 (0.87)	8.12 (0.51)
Lockup Period	1.77 (1.00)	8.86 (0.50)	37.40 (6.06)	62.60 (6.06)	-6.26 (1.17)	10.08 (0.58)
No Notice Period	-0.62 (1.45)	9.72 (0.56)	56.50 (6.22)	43.50 (6.22)	-9.48 (1.22)	8.72 (1.11)
Notice Period	0.73 (0.87)	9.03 (0.41)	43.67 (5.63)	56.33 (5.63)	-7.72 (0.90)	8.71 (0.49)

question is whether these costs are sufficiently large to match the heterogeneity observed in the data.

5.4. Return decomposition over time

5.4.1. Hedge funds versus mutual funds

The universe of hedge funds has expanded substantially since 1994. As a result, the average return decomposition may be subject to notable time trends. Using the CP model, we track the evolution of (i) the average alpha component $\hat{M}_{1,ac}^k = \frac{1}{n_x} \sum_i^{n_x} \hat{a}c_i^k$, (ii) the average beta component due to the market $\hat{M}_{1,bc,m}^k = \frac{1}{n_x} \sum_i^{n_x} \hat{b}c_{i,m}^k$, and (iii) the average beta component due to the non-market (nm) factors (all factors but the market) $\hat{M}_{1,bc,Inm}^k = \frac{1}{n_x} \sum_i^{n_x} \hat{b}c_{i,Inm}^k$. We start the analysis at the end of 2003 and estimate each cross-sectional average using the entire return history for each fund up to that point in time. As we move forward in time, we expand the set of return observations and add new hedge funds once they satisfy the fund selection rule. For comparison purposes, we conduct the same analysis for mutual funds using the traditional Carhart model.

Fig. 3 identifies two sources of convergence between hedge funds and mutual funds. First, performance becomes increasingly similar—at the end of 2020, the gap in average alphas drops to 1.6%. Understanding the drivers of this trend is an important topic for hedge fund researchers. One intuitive explanation is the presence of scalability constraints. As a result of the growth of the hedge fund industry, it becomes increasingly difficult to maintain the same performance level. Bollen et al. (2021) find support for this explanation, but also suggest that central bank interventions might have reduced the profitability of hedge fund strategies. Second, hedge funds load increasingly on the equity market. At the end of 2020, $\hat{M}_{1,bc,m}^k$ reaches its highest level at 2.3% per year. Whereas it remains smaller than for mutual funds (7.8%), it follows the same trend after the 2008 crisis. This trend is relevant for investors eager to capture hedge fund alphas. Given the sizable market exposure of hedge funds, they need a proper hedging strategy to maintain their optimal allocation to the equity market.

5.4.2. Differences between models

Next, we examine the differences between models over time. We compute the difference in average alphas between (i) the CAPM and the Fung-Hsieh model, (ii) the CAPM and the CP model, (iii) the Fung-Hsieh and CP models. Fig. 4 reveals that the CP model always delivers smaller alphas as hedge funds consistently use non-market factors to boost their returns. The average contribution of these factors captured by $\hat{M}_{1,bc,Inm}^k$ represents at least 52% of the average hedge fund return between 2003 and 2020. We also see that the Fung-Hsieh model does a better job at capturing returns than the CAPM early on. However, this ability progressively vanishes after the 2008 crisis.

The gap produced by the CP model slightly narrows over time as $\hat{M}_{1,bc,Inm}^k$ falls from 4.4% to 2.9% per year. We find little evidence that hedge funds reduce their factor exposures over time. Instead, the decrease in $\hat{M}_{1,bc,Inm}^k$ is caused by the reduction in alternative factor premia.²² An open question is whether this trend will continue. On the one hand, variance, carry, and TS momentum have high Sharpe ratios—an observation made by Dew-Becker et al. (2017), Kojien et al. (2018), and Moskowitz et al. (2012). As a result, investors should have incentives to exploit these factors and drive down their returns. On the other hand, Ilmanen et al. (2021) go back to 1926 and find little evidence of arbitrage activities that permanently reduce alternative factor premia.

The large difference between the Fung-Hsieh and CP models may be surprising as one might expect the straddles to overlap with the variance factor. Conceptually, there are two reasons why this is not the case. First, a straddle position benefits from trends but not necessarily from variance. For example, consider a highly volatile period for an asset during which the ending price is the same as the starting price. In this case, the payoff of the straddle is zero, whereas the payoff of the variance factor is large. Second, the straddles in the Fung-Hsieh model cover the bond, currency, and commodity markets. In contrast, the variance factor focuses on the equity market.

²² For instance, the premium on carry drops from 8.8% to 6.8% per year between 2003 and 2020.



Fig. 3. Time-Variation in the Average Return Components.

This figure compares the return decomposition for hedge funds and mutual funds. Panel A plots the evolution of the average return components (annualized) under the CP model: (i) the alpha component (Alpha), (ii) the beta component due to the market (Beta Mkt), and the beta component due to the non-market factors (Beta Non-Mkt), which are size, illiquidity, BAB, variance, carry, and TS momentum. At the end of each month, we measure the three components of each fund using its entire return history up to that point and take cross-sectional averages across all existing funds. The initial estimates cover the period 1994 to 2003, while the last ones cover the entire period 1994 to 2020. Panel B repeats the analysis for mutual funds using the traditional Carhart (1997) model, which includes size, value, and momentum as non-market factors.

5.5. The sophistication of hedge fund investors

5.5.1. Impact of sophistication on hedge fund valuation

In addition to sharpening the estimation of the distributions ϕ_{ac}^* and ϕ_{bc}^* , our comparison analysis measures how hypothetical investors with different sophistication levels value hedge funds. We first consider a sophisticated investor able to replicate all the factors in the CP model. The valuation of this investor is given by the alpha under the CP model. We can formalize this intuition using the stochastic discount factor (SDF) framework. Writing the investor SDF m_t^k as a linear function of the CP factors, we have $\alpha_i^k = (1 + r_f)E[m_t^k r_{i,t}]$, where r_f is the risk-free rate. A positive α_i^k signals that the investor can increase his overall utility by investing in the fund (e.g., Chen and Knez, 1996; Ferson, 2013).²³ Next,

we consider a less sophisticated investor who can only invest in the equity market. Her hedge fund valuation is given by the CAPM alpha: $\alpha_i^0 = (1 + r_f)E[m_t^0 r_{i,t}]$, where m_t^0 is a linear function of the market.

Table 6 shows that the average valuation is close to zero for the CP investor (0.4% per year), but substantially higher for the CAPM investor (2.9%). This valuation gap is consistent with intuition. Whereas both investors value the alpha component ac_i^* , the CAPM investor also values the beta component due to all the strategies she cannot replicate. This component, denoted by $bc_{i,O}^*$ in Equation (5), includes the contributions of the TS momentum, variance, and carry factors.²⁴

²³ See also Almeida et al. (2020) and Karehnke and de Roon (2020) for a recent application of the SDF framework in which investors have nonlinear preferences (i.e., m_t is a nonlinear function of the factors).

²⁴ This point is well summarized by Cochrane (2011): “I tried telling a hedge fund manager, ‘You don’t have alpha. Your returns can be replicated with a value-growth, momentum, currency and term carry, and short-vol strategy.’ He said, ‘Exotic beta is my alpha. I understand those systematic factors and know how to trade them. My clients don’t.’”

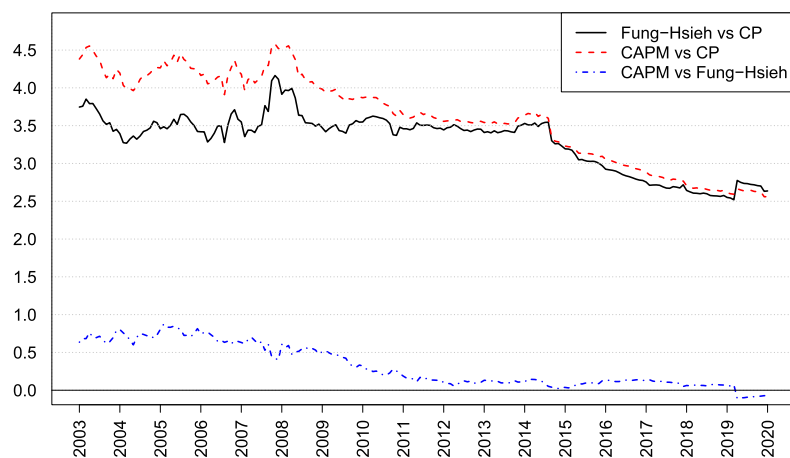


Fig. 4. Time-Variation in the Differences between Models.

This figure plots the evolution of the differences in average alphas (annualized) between: (i) the CAPM and the CP model, and (ii) the Fung-Hsieh model and the CP model. At the end of each month, we measure the alpha of each fund under all three models, take cross-sectional averages across all existing funds, and then compute the differences between models. The initial estimates cover the period 1994 to 2003, while the last ones cover the entire period 1994 to 2020. The differences in the average beta components are not reported because they are identical to the ones obtained with the alpha component (with opposite signs).

5.5.2. Measuring sophistication using flows

As a final exercise, we study the sophistication of real-world hedge fund investors. Our analysis builds on the premise that investors learn about funds by observing past returns. As they update their valuation over time, they reallocate capital accordingly. If this mechanism is at play, fund flows contain information about investor preference for alpha and beta. With misspecified models, the learning process is likely to be noisier than for mutual funds. To address this issue, we compare the entire distributions of the alpha and beta components between low- and high-flow funds.

We proceed in three steps. First, we follow Barras et al. (2010) and partition our data into non-overlapping subperiods of five years, beginning with 1996 to 2000 and ending with 2016 to 2020. For each subperiod, we include all funds that pass the fund selection rule and compute their average monthly flows and return decomposition obtained with the CP model ($\hat{a}c_i^k$, $\hat{b}c_{i,m}^k$, $\hat{b}c_{i,I_{nm}}^k$). Second, we sort funds into flow quintiles (from low to high) and pool these five-year records together across all time periods. Third, we apply our methodology to compute the distributions of the alpha and beta components for each (pooled) flow quintile.

Panel A of Table 11 reveals that flows are primarily directed into funds with positive contemporaneous alphas. In the high-flow group, the alpha is equal to 3.3% per year on average and positive for 70.4% of the funds.²⁵ In the low-flow group, these numbers drop substantially (-0.6% and 45.2%). In spite of these differences, the evidence is consistent with our premise that learning about fund alphas is quite noisy—we observe a large overlap between the alpha distributions.

If real-world investors are unsophisticated, they not only chase alphas, but also past or market-adjusted returns. In other words, they also direct capital into funds that load aggressively on the market and alternative factors (*i.e.*, funds with high $\hat{b}c_{i,m}^k$ and $\hat{b}c_{i,I_{nm}}^k$). Panels B and C do not support this interpretation. For the two sets of factors, the average beta components are actually larger in the low-flow group than in the high-flow group. In short, the overall evidence suggests that real-world investors chase alphas more aggressively than betas and are therefore closer to the CP investor than the CAPM investor in terms of sophistication.

²⁵ If the hedge fund industry is subject to capacity constraints, the cumulative flows into positive-alpha funds could explain why the average hedge fund alpha decreases over the period 2003–2020 (see Fig. 3).

6. Conclusion

Decomposing hedge fund returns is challenging because models are likely misspecified—that is, they omit relevant factors for capturing hedge fund strategies. Model misspecification makes the estimation of the alpha and beta components both imperfect and noisy. To mitigate these challenges, we develop a new approach to estimate and compare the distributions of the alpha and beta components across models. Our approach improves the imperfect separation between alpha and beta contributions by identifying models less prone to misspecification. It also explicitly accounts for estimation noise based on a full-fledged asymptotic theory in a large cross-section of funds.

Our comparison analysis yields several insights. We find that the standard models are similar to the CAPM and thus ill-equipped for separating alpha and beta. In contrast, several economically motivated factors—primarily, TS momentum, variance, and carry—capture the returns of hedge funds in all investment categories. Including these factors increases the relative importance of the beta component and uncovers a gradual convergence in performance towards mutual funds. Another important finding is the large heterogeneity observed across funds. This result has several implications for the use of benchmarks based on style information, the role of uncertainty in the fund selection process, and the predictions of equilibrium models of active management.

Our methodology is flexible and can be applied in other situations where models are misspecified. For instance, it can be used to further improve the decomposition of hedge fund returns in more specialized investment categories. It can also be applied to international mutual funds for which the set of trading strategies is substantially larger than for traditional US equity funds.

CRedit authorship contribution statement

David Ardia: Data curation, Formal analysis, Investigation, Software, Writing – review & editing. **Laurent Barras:** Conceptualization, Investigation, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Patrick Gagliardini:** Conceptualization, Methodology. **Olivier Scaillet:** Conceptualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 11**Fund Flows and Investor Sophistication.**

This table measures how investor flows contemporaneously respond to the different components of the average fund return under the CP model: (i) the alpha component, (ii) the beta component due to the market, and (iii) the beta component due to the non-market factors (size, illiquidity, BAB, variance, carry, TS momentum). For each of the non-overlapping five-year periods between 1996 and 2020, we measure the three components for all funds. We then rank them according to their average monthly net flows and group them into quintiles (low, 2, 3, 4, high). Panel A reports the characteristics of the cross-sectional distribution of the alpha components (pooled over all five-year periods) for the five flow groups. We report the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panels B and C repeat the analysis for the beta components due to the market and the non-market factors.

Panel A: Distribution of the Alpha Components						
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low	-0.57 (1.54)	10.71 (0.79)	54.84 (5.88)	45.16 (5.88)	-10.53 (1.33)	9.55 (0.74)
2	0.39 (1.33)	9.03 (0.72)	47.82 (5.66)	52.18 (5.66)	-9.01 (0.98)	9.15 (0.83)
3	1.52 (1.32)	8.42 (0.76)	41.49 (6.10)	58.51 (6.10)	-7.18 (0.90)	10.33 (0.65)
4	2.92 (1.44)	8.85 (0.82)	34.57 (6.90)	65.43 (6.90)	-5.97 (1.18)	13.00 (0.56)
High	3.25 (1.24)	9.87 (0.61)	29.61 (5.31)	70.39 (5.31)	-4.92 (0.98)	13.18 (0.74)

Panel B: Distribution of the Beta Components (Market Factor)						
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low	2.32 (1.98)	4.19 (1.75)	18.19 (19.41)	81.81 (19.41)	-0.53 (0.84)	8.37 (3.00)
2	3.15 (2.45)	5.09 (1.75)	14.25 (18.87)	85.75 (18.87)	-0.17 (1.19)	10.34 (4.27)
3	2.91 (2.26)	4.94 (1.84)	15.85 (18.60)	84.15 (18.60)	-0.17 (1.13)	9.48 (4.06)
4	2.40 (1.84)	4.42 (1.74)	19.96 (15.33)	80.04 (15.33)	-0.33 (0.71)	8.29 (3.30)
High	1.95 (1.45)	3.90 (1.61)	23.07 (13.06)	76.93 (13.06)	-0.46 (0.45)	7.31 (2.63)

Panel C: Distribution of the Beta Components (Non-Market Factors)						
	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low	2.38 (0.90)	8.18 (3.58)	32.13 (8.54)	67.87 (8.54)	-4.43 (1.93)	10.44 (3.66)
2	2.13 (0.83)	7.36 (3.03)	35.93 (12.43)	64.07 (12.43)	-4.26 (1.59)	9.60 (3.57)
3	2.14 (0.81)	6.94 (3.28)	32.07 (11.95)	67.93 (11.95)	-4.23 (2.02)	9.34 (3.59)
4	2.30 (0.78)	7.16 (3.19)	34.30 (8.29)	65.70 (8.29)	-3.88 (1.79)	9.63 (3.59)
High	2.37 (0.81)	7.82 (3.09)	33.70 (6.23)	66.30 (6.23)	-3.81 (1.70)	8.65 (3.46)

Data availability

The authors do not have permission to share data.

Code and fake pseudodata are available at <https://doi.org/10.5281/zenodo.10459612>.

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Appendix A. Supplementary material

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